

SAMPLING DISTRIBUTIONS

9.1 Introduction

Sampling is often used in daily life without being noticed. For example, a jeweller tests the purity of gold by examining a small portion instead of melting the entire ornament. A shopkeeper tastes a few cashews before deciding the quality of the whole sack. A farmer judges the sweetness of a mango tree by plucking and tasting only one or two fruits. These are everyday situations where decisions are based on a sample rather than the entire population.

A doctor studies a few cells from a tissue sample to diagnose a disease in the whole body. A factory supervisor checks only a few bulbs from a large batch to decide on the quality of production. Thus, in most practical cases, complete examination is not possible, and decisions are made from selected samples.

In this chapter, we will study **Sampling**, which deals with different ways of selecting a part of the population, and **Sampling Distribution**, which explains how the results obtained from different samples behave. Together, these concepts form the basis of **statistical inference** and help in drawing reliable conclusions about the population.

9.2 Concept of Population

In statistics, the term population does not always mean people. It refers to the entire group of items or individuals about which information is required. A population can be large or small, finite or infinite. Since it contains every possible unit of study, it becomes the complete source of data for statistical analysis.



Meaning : Population

A population refers to the entire set of individuals, objects, or events that a study wants to understand and describe. It includes every member of the group.

- Population is complete.
- It may be finite (like 2000 students in a college) or infinite (like bacteria in a glass of water).
- Parameters like mean (μ) and standard deviation (σ) belong to the population.

Population = Full set of data that the researcher wants to understand.

Example 1 Students in a University

Suppose the Registrar of Bangalore University wants to know the average attendance percentage of all undergraduate students.

- Population = All undergraduate students enrolled in Bangalore University.
- If there are 40,000 such students, then the population size is 40,000.
- It is not practical to check every student's attendance manually.

Example 2 Manufacturing Company

A factory produces 15,000 ceiling fans every month. The quality control department wants to know the average weight of these fans.

- Population = All 15,000 fans produced that month.
- It is difficult to weigh each fan, so they study only a small sample.
- The entire 15,000 fans form the population.

Example 3 Voter Survey

A news channel wants to conduct a voter opinion poll in Karnataka before elections.

- Population = All eligible voters in Karnataka (around 5 crore people).
- It is impossible to ask all voters, so they take a sample.
- The entire set of voters is the population.

9.2.1 Finite Population and Infinite Population

A **finite population** is one that contains a countable number of units. In such cases, the total number of elements can be determined exactly. This total is called the population size.

Examples:

- The number of books in a college library.
- The number of students in a classroom.
- The number of vehicles registered in a city in one year.
- The number of mobile phones produced in a factory on a given day.

An **infinite population** is one where the number of units is too large to be counted or theoretically endless. In these cases, it is impossible to know the exact population size.

Examples:

- The number of grains of sand on a seashore.
- The number of raindrops falling during a heavy shower.
- The number of stars in the universe.
- The number of possible outcomes when tossing a coin repeatedly.

9.2.2 Census Method

Information about a population can be collected in two main ways: the **census method** and the **sample method**. In the census method, every element of the population is studied without leaving out any unit. This means complete information is collected from the entire population.

Example: If the aim is to study the average annual income of families in a village with 1000 families, then the income of all 1000 families must be collected. Similarly, when the government conducts a national population census, details of every household are included. Since no element is left out, the census method provides complete and accurate information, but it is often time-consuming and costly.

Merits of Census Method

1. Data is collected from every unit of the population without leaving anything out.
2. The results are accurate and reliable since the whole population is studied.
3. It allows for a detailed and intensive study of the subject.
4. The information collected can be used for different types of surveys and analyses.
5. It is very useful for studying rare events, such as the area under a particular crop, the yield of crops, the number of people in a specific age group, or their level of education. This is why governments across the world conduct a population census every 10 years.
6. The complete data collected through census serves as a strong base for future surveys and research.

Limitations of Census Method

1. It requires a large number of people to collect data, which makes it an expensive method.
2. It involves high cost of money, labour, time, and effort.
3. It is not practical for infinite populations, such as stars in the sky or grains of sand.
4. It becomes difficult to organize and manage when the population is very large.
5. Sometimes the data collected may lose relevance by the time it is processed, because the method takes too long.

9.3 Concept of Sample

In statistics, studying the entire population is often not possible due to large size, cost, or time limitations. Instead, a sample is taken. A sample is a small part of the population selected in such a way that it represents the entire population.

Meaning : Sample

A sample refers to a subset of the population chosen for study. It provides the basis for drawing conclusions about the entire group.

- Sample is a part of the population.
- It should be representative of the whole population.
- Statistics like sample mean ($\bar{x}$) and sample standard deviation (s) are calculated from the sample.
- By studying a sample, we can make predictions about the population.

Sample = A small group taken from the population to study the whole.

Example 1 Students in a University

Suppose the Registrar of Bangalore University wants to know the average attendance percentage of all undergraduate students.

- Population = All undergraduate students enrolled in Bangalore University.
- Sample = A group of 500 students randomly chosen from different colleges.
- The average attendance of this sample is used to estimate the attendance of all 40,000 students.

Example 2 Manufacturing Company

A factory produces 15,000 ceiling fans every month. The quality control department wants to know the average weight of these fans.

- Population = All 15,000 fans produced that month.
- Sample = 200 fans selected randomly from the production line.
- Testing these 200 fans gives an estimate of the average weight of all fans.

Example 3 Voter Survey

A news channel wants to conduct a voter opinion poll in Karnataka before elections.

- Population = All eligible voters in Karnataka (around 5 crore people).
- Sample = 10,000 voters selected from different districts.
- The opinions of these 10,000 voters are used to predict the overall voting trend.

9.3.1 Sampling

In statistics, it is usually not possible to study the whole population because of its large size, high cost, and time limitations. Instead, a small part of the population is selected for study. This process is called sampling.

**Meaning : Sampling**

Sampling is the method of selecting certain units (like households, consumers, companies, or students) from a population for analysis. In simple words, the process of choosing a sample is called **sampling**.

- The smallest non-divisible part of the population is called a **unit**. It must be clearly defined so that there is no confusion.

Example : If a unit is defined as a household, then it should be specified in such a way that one person cannot be counted in two households and no person belonging to the population is left out.

- A sample is a finite subset of the population.

Example: Out of 10,000 employees in a company, if 500 employees are selected for a job satisfaction survey, then these 500 employees form a sample.

- The number of units in the sample is called the sample size.

Example: In the above case, the 500 employees chosen for the survey represent the sample size of 500.

9.3.2 Principle of Statistical Regularity

The theory of sampling is based on the **principle of statistical regularity**. This principle says that "If a sample is chosen randomly and has a moderately large number of items, then the characteristics of that sample will be almost the same as those of the population". In other words, a well-chosen random sample can represent the population.

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Example : Suppose a school has 2,000 students, and the average height of all students (population) is 150 cm. If we randomly select 200 students (sample), then the average height of those 200 students will be very close to 150 cm. That is why, instead of studying all 2,000 students, we can study just 200 students and still get almost the same result.

9.3.3 Need for Sampling

In many practical situations, it is not possible or useful to study the entire population. For example, in a factory, the raw materials department receives goods in large lots. Before these goods are used in production, the inspection department checks their quality.

There are two possible approaches:

1. Check every item in the lot (complete enumeration or census).
2. Select only a part of the lot (a sample) and inspect it, then use the result to judge the quality of the whole lot.

The first method (census) is often not practical because:

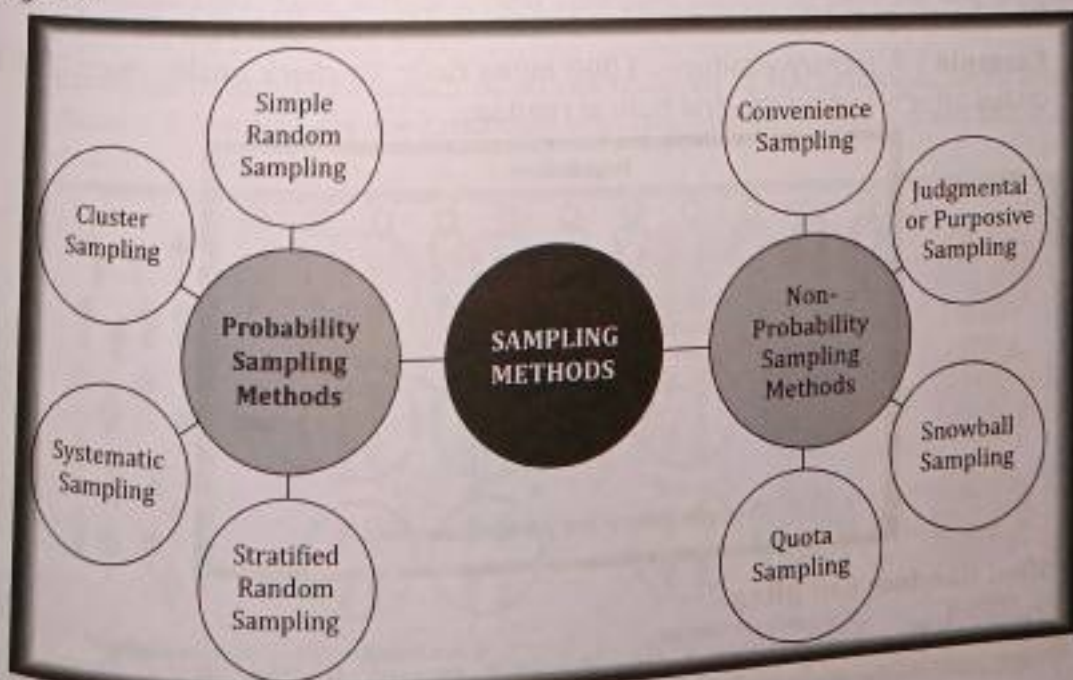
- It requires a lot of time.
- It involves high cost.

The second method (sampling) has clear advantages:

- It is less expensive.
- It takes less time while still giving reliable results.

9.3.4 Methods of Sampling (or) Sampling Methods

When studying a large population, it is often not possible to collect data from every individual. In such cases, a smaller group called a sample is chosen to represent the whole population. The method used to select this sample is called a sampling method. Sampling methods are broadly divided into two categories:



A. Probability Sampling Methods

Probability sampling means choosing a sample in such a way that every person or item in the whole population has a fair and known chance of being selected.

Think of it like a lucky draw. Everyone's name is written on a slip and put in a box. If one slip is drawn, each person had an equal chance to be chosen. This is what probability sampling does—it removes bias and ensures fairness.

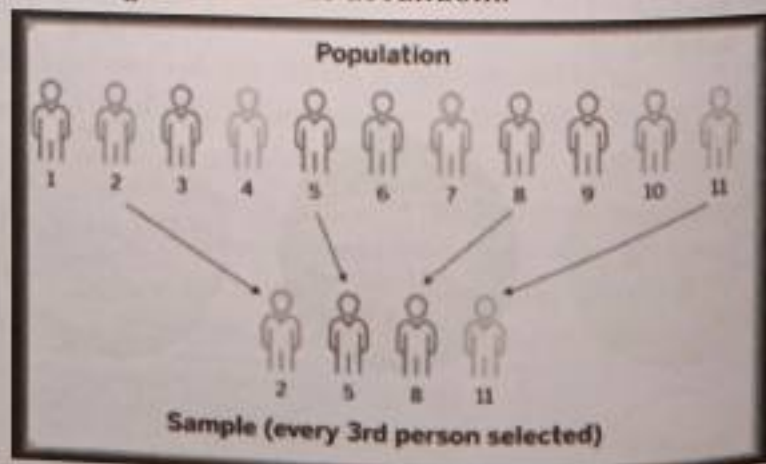
1. Simple Random Sampling

- **Definition** : Every item in the population has an equal chance of being selected.
- **How it Works** : Selection can be done through lottery method, random number tables, or computer-generated random numbers.
- **Advantages** : Very simple, unbiased, and accurate when population is homogeneous.
- **Disadvantages** : Difficult when population is large and scattered.
- **Example** : From a list of 500 employees, select 50 names by using a lottery system. Each employee has equal chance of being selected.



2. Systematic Sampling

- **Definition** : Items are chosen at regular intervals from an ordered list. The first item is selected randomly, and then every n th item is included.
- **How it Works** : Suppose population size is N , sample size required is n . Then the interval is $k = \frac{N}{n}$. A random starting point between 1 and k is chosen, and then every k th element is selected.
- **Advantages** : Easy to apply, less time-consuming.
- **Disadvantages** : If there is a hidden pattern in data, results may be biased.
- **Example** : A factory produces 1,000 bulbs daily. To check quality, every 20th bulb is tested after selecting the first bulb at random.

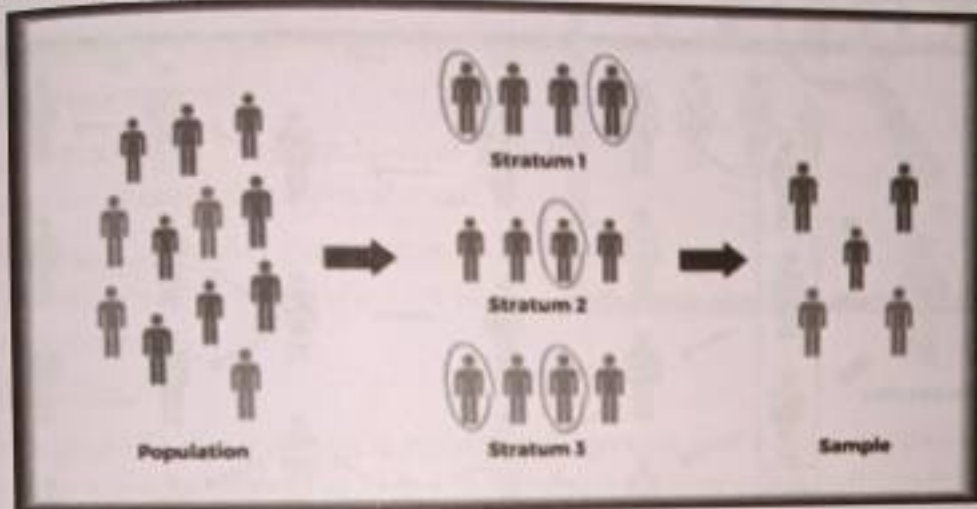


3. Stratified Random Sampling

- **Definition** : Population is divided into groups (called strata) based on certain characteristics, and random samples are drawn from each stratum.

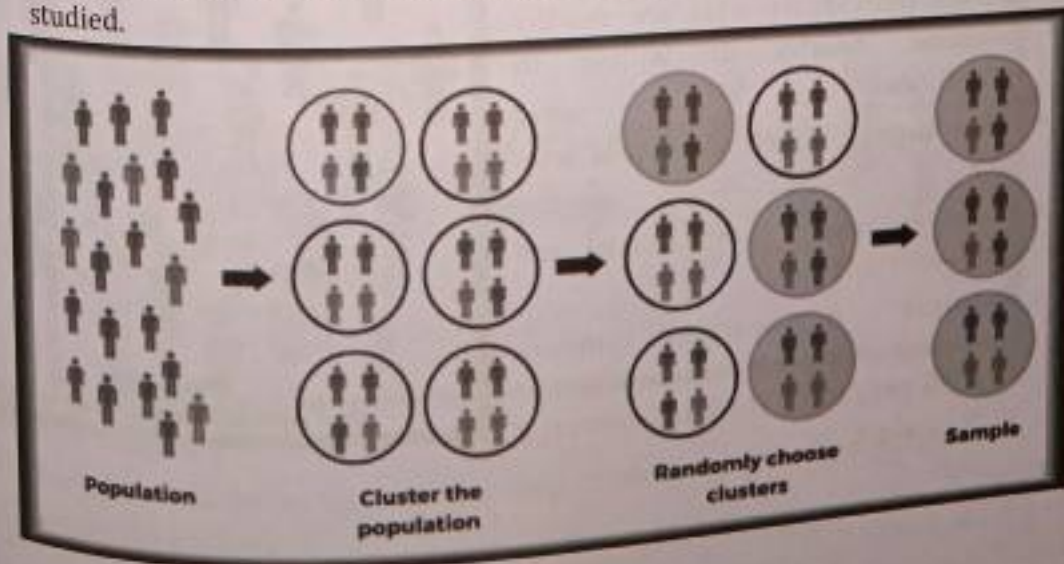
4. Clus

- **How it Works** : The groups are homogeneous internally but heterogeneous when compared with each other. Samples are taken proportionately from each stratum.
- **Advantages** : Ensures representation of all groups. Gives more accurate results when population is diverse.
- **Disadvantages** : Requires detailed knowledge of population characteristics.
- **Example** : A college wants to survey students' study habits. Students are divided into strata based on streams: Science, Arts, and Commerce. Then random samples are taken from each stream.



4. Cluster Sampling

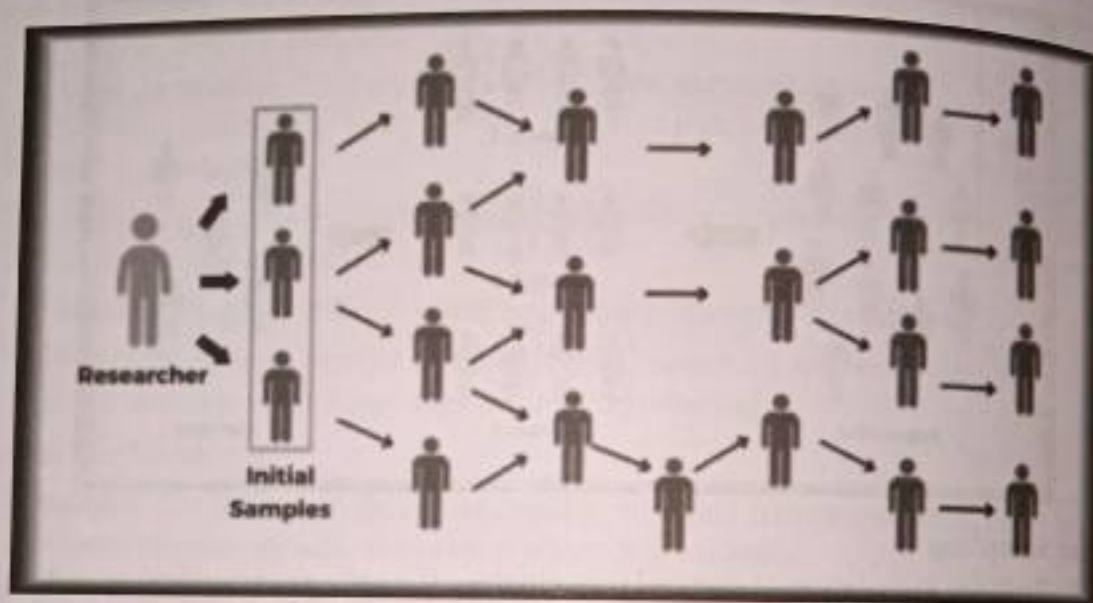
- **Definition**: Population is divided into clusters (usually geographical or natural groups). Some clusters are randomly selected, and all units within those clusters are studied.
- **How it Works**: Instead of selecting individuals, entire groups are selected.
- **Advantages**: Saves time and cost when population is very large and spread out.
- **Disadvantages**: Less accurate than stratified sampling, because not all clusters may represent the population well.
- **Example**: To study the health of school children in Karnataka, schools are treated as clusters. A few schools are randomly selected, and all students from those schools are studied.



B. Non-Probability Sampling Methods

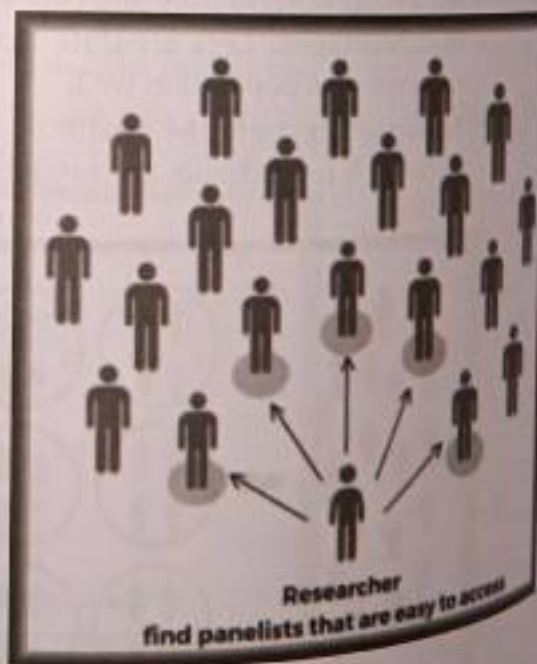
Non-probability sampling means choosing a sample where every person or item in the population does not have a fair or equal chance of being selected. The selection is based on convenience, judgement, or other non-random methods.

Think of it like a teacher choosing students for a quiz team. Instead of giving every student an equal chance, the teacher may pick only the top scorers or the students sitting in the first row. Here, not everyone had a chance to be selected.



1. Convenience Sampling

- **Definition** : Samples are chosen from members of the population who are easiest to access.
- **How it Works** : The investigator selects individuals who are readily available, nearby, or willing to participate.
- **Advantages** : Quick, easy, and low-cost.
- **Disadvantages** : Highly biased and often unreliable.
- **Example** : A researcher standing outside a shopping mall asking the first 50 people for their opinion on a new product.



2. Judgement

3. Quota

- Definition
- How it works
- Advantages
- Disadvantages
- Example

4. Snowball

- Definition
- How it works

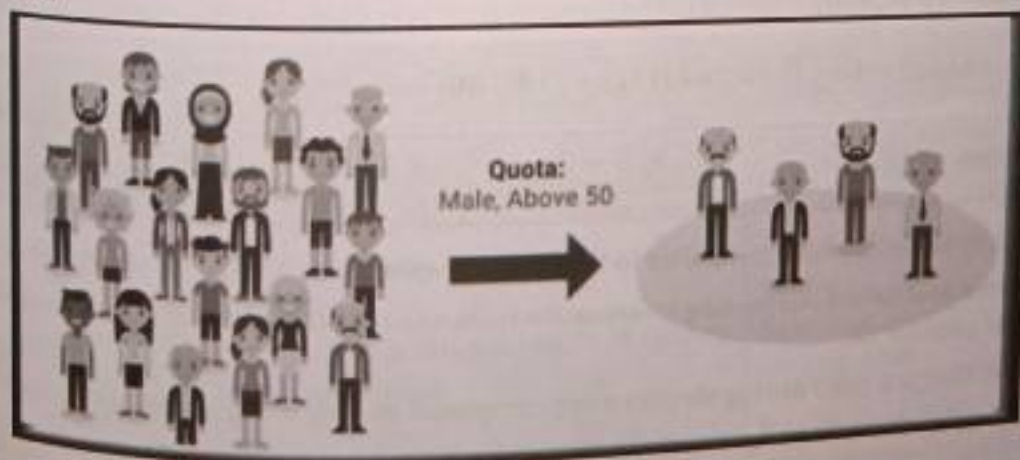
2. Judgemental or Purposive Sampling

- **Definition :** The investigator uses their own judgement to select individuals who are most suitable or representative for the study.
- **How it Works :** Based on experience or expertise, the investigator decides who should be included in the sample.
- **Advantages :** Useful when specific expertise is needed.
- **Disadvantages :** Very subjective and may not represent the population.
- **Example :** A teacher choosing only the best students to study "effective study techniques."



3. Quota Sampling

- **Definition :** The population is divided into groups, and a fixed number (quota) is taken from each group. Within each group, selection is done using non-random methods such as convenience.
- **How it Works :** Similar to stratified sampling, but selection inside each group is not random.
- **Advantages :** Ensures representation of different categories.
- **Disadvantages :** Still prone to bias because the choice inside each group is not random.
- **Example :** In a political survey, a researcher is asked to collect responses from 100 men and 100 women. They approach the first available men and women until the quota is filled.



4. Snowball Sampling

- **Definition :** Existing participants help in identifying further participants. The sample grows like a snowball.

- **How it Works** : Used when the population is hard to locate. One participant refers another, and the chain continues.
- **Advantages** : Useful for hidden or hard-to-reach populations.
- **Disadvantages** : Not representative of the whole population, depends on referrals.
- **Example** : A researcher studying drug users asks one user for contacts of other users, who then provide more contacts.

9.3.5

Principles of Sampling

For sampling to give good results, certain principles must be followed. These principles ensure that the sample gives a true picture of the population.

1. **Principle of Statistical Regularity** : If a moderately large number of units are selected at random from a population, they are almost certain to represent the characteristics of the whole group.

Example: If 200 students are chosen at random from 40,000 students, their average attendance will be close to the overall average.

2. **Principle of Inertia of Large Numbers** : As the size of the sample increases, the results become more accurate and reliable, provided other conditions remain the same.

Example: A survey of 1,000 voters is more reliable than a survey of only 100 voters.

3. **Principle of Validity** : The method of sampling should provide results that are valid and useful for estimating the parameters of the population.

Example: If income data is collected from a properly chosen sample of families, it can be used to estimate the income level of the whole community.

4. **Principle of Optimisation** : Sampling should be planned in such a way that maximum information is obtained with minimum cost, time, and effort.

Example: Testing 300 bulbs out of 30,000 should give reliable results about quality while saving cost and labour.

9.3.6

Advantages and Disadvantages of Sampling



Advantages of Sampling

Sampling has many benefits when compared to the census method:

1. **Saves Time and Effort** : Collecting information from a sample takes less time and labour than studying the whole population.

Example: Opinion polls during elections are completed quickly using a small sample of voters.

2. **Reduces Cost** : Sampling requires fewer resources in terms of money, staff, and man-hours.

Example: A company tests 200 bulbs instead of all 20,000, saving testing costs.

3. **Can Provide Accurate Results** : A carefully chosen sample can give results that are almost as accurate as a census, especially when modern statistical methods are applied.

4. **Practical for Large or Infinite Populations** : When the population is very large, infinite, or cannot be fully studied, sampling becomes the only possible method.

Example: To test water quality in a river, only water samples are collected.

5. **Useful in Destructive Testing** : When testing destroys the item, census is impossible, and sampling is the only option.

Example: Testing the breaking strength of steel rods.

6. **Flexibility and Adaptability** : Sampling can be applied to different fields such as business, medicine, agriculture, and social sciences with equal success.

7. **Scope for Detailed Study** : Since fewer items are studied, more resources can be spent on in-depth analysis of the sample units.

Example: A doctor can study 200 patients thoroughly instead of trying to study all 10,000 patients.

Disadvantages of Sampling

Despite its advantages, sampling has some limitations:

1. **Requires Expertise** : The method of sampling must be planned and executed by trained and experienced persons. If not, the results will not be reliable.

2. **Chance of Bias** : If the sample is not chosen properly, it may not represent the population, leading to misleading conclusions.

Example: A survey done only in urban areas may ignore rural opinions.

3. **Sampling Errors** : Since only a part of the population is studied, there is always a possibility of error. Census, in theory, avoids this problem.

4. **May Miss Rare Cases** : Rare events or unusual items may not be included in the sample.

Example: In medical research, rare side effects of a drug may not appear in a small sample.

5. **Not Suitable for Small Populations** : When the population size is very small, census is more practical and accurate than sampling.

6. **Less Accuracy Compared to Census** : Even a good sample cannot match the 100% accuracy of a census, especially when very high precision is required (Example, national income estimates).

9.3.7 Difference between Census Method & Sampling Method

Parameters	Census method	Sampling method
1. Nature of enquiry	The extensive enquiry is conducted at each and every unit of the population is studied.	The limited enquiry is conducted as only a few units of the population are studied.
2. Economy	It requires a large amount of money, time and labour.	Relatively less money, time and labour are required.
3. Suitability	It is more suitable if the population is heterogeneous in nature	It is more suitable if the population is homogeneous in nature.

4. Reliability and accuracy	Results are quite reliable and accurate under the Census method.	The understanding method results are less reliable because a high degree of accuracy is not achieved.
5. Organization and supervision	It is very difficult to organize and supervise the census method.	The sampling method is comparatively easy to organize and supervise.
6. Verification	Under this method results of the investigation cannot be verified.	Under this method, results can be tested taking out another sample.
7. Nature of method	Census method is an old method of investigation and it is not a very scientific method.	The sampling method is a new and practicable method, it is a scientific method.

9.3.8 Sampling with and without Replacement

When selecting a sample from a population, two approaches are possible: with replacement and without replacement.

1. Sampling with Replacement :

In this method, once an element is selected, it is returned back to the population before the next draw.

- This means the same element can be chosen more than once.
- Every time a selection is made, all elements of the population have the same chance of being selected.

If the population has N elements and a sample of size n is to be drawn, then the number of possible samples is: N^n

Example : If we have four numbers 3, 4, 6, 7, constitute a population. We have to choose a sample of 2 elements from this population.

Samples with replacement = [3,3], [3,4], [3,6], [3,7], [4,3], [4,4], [4,6], [4,7], [6,3], [6,4], [6,6], [6,7], [7,3], [7,4], [7,6], [7,7]
 $= 16 = (4)^2$

2. Sampling without Replacement

In this method, once an element is selected, it is not returned to the population.

- This means the same element cannot appear more than once in the same sample.
- After one element is chosen, the next selection is made from the remaining elements.

In a population containing N observations, the number of samples each containing n observations = ${}^N C_n$ if the sampling is done with replacement.

Example : If four numbers 3, 4, 6, 7, constitute a population. We have to choose a sample of 2 elements from this population.

Samples with replacement = [3,4], [3,6], [3,7], [4,6], [4,7], [6,7]
 $= 6 = {}^4 C_2$

9.4 Population Statistics

1. A parameter
2. A statistic

9.4.1 Parameter

A parameter is a statistical measure that tells us something about the population. It is a fixed value and does not change. It is used to estimate the population parameter.

• Parameter

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1. Population

2. Population

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4. Population

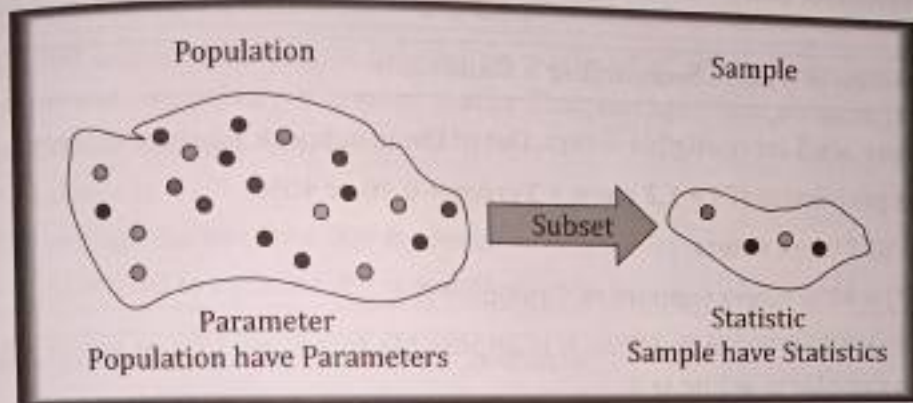
5. Population

9.4 Population Parameters and Sample Statistics

In statistics, two important measures are used: parameters and statistics.

1. A parameter describes the entire population.
2. A statistic describes a sample taken from that population.

Inferential statistics is the process of using sample statistics to estimate population parameters.



9.4.1 Parameter

A parameter is a numerical value that gives information about a certain feature of the entire population. It tells us something about the whole group, not just a part of it. Parameters are often unknown because it is not possible to collect data from every member of a large population. Instead, we try to estimate them using sample statistics.

A parameter is like the real truth about a population. It is fixed, but usually hidden. We use samples to estimate it.

- **Parameter** is a descriptive measure of a population.
- Parameters are fixed values for a given population.

Example: If we consider all students of a school, then their average height (μ) is a fixed number for that school.

- Since collecting data from the whole population is difficult, parameters are mostly unknown in practice.
- Parameters help us describe the true picture of the population, while samples only give estimates.
- A parameter is usually denoted with letters of the lower case Greek alphabet such as mean μ and standard deviation σ .

• Examples of Parameters :

1. Population Mean $\rightarrow \mu$ (mu)
2. Population Standard Deviation $\rightarrow \sigma$ (sigma)
3. Population Variance $\rightarrow \sigma^2$
4. Population Proportion $\rightarrow P$
5. Population Median $\rightarrow M$

Example 1 Average Marks of Students

Suppose a university has 50,000 students. If the university calculates the average marks of all 50,000 students, the result is a population mean (μ).

- Population = 50,000 students
- Parameter (μ) = true average marks of all students
- This value is fixed for that year. Even if it is difficult to calculate, it still exists as a parameter.

Example 2 Proportion of Voters Supporting a Candidate

In a state election, there are 3 crore eligible voters. Out of them, suppose 1.2 crore voters support Candidate A. Then, the population proportion (P) = $1.2 \text{ crore} \div 3 \text{ crore} = 0.40$ or 40%.

- Population = all 3 crore voters
- Parameter (P) = 40% voters supporting Candidate A
- This is the true value, but in practice, it is impossible to ask all 3 crore voters. That is why survey agencies take samples to estimate it.

Example 3 Defective Rate in a Factory

A factory produces 2,00,000 mobile phones in a month. Suppose 4,000 phones are defective.

Then, the defective proportion (P) = $4,000 \div 2,00,000 = 0.02$ or 2%.

- Population = all 2,00,000 phones
- Parameter (P) = true defective rate = 2%
- The management may not test all phones. Instead, they test a sample. But the 2% defective rate is the actual population parameter.

9.4.2

Statistic

A statistic is a numerical value that gives information about a certain feature of a sample (a subset of the population). It tells us something about the part of the group that has been studied, and from it we try to make inferences about the entire population.

Unlike parameters, statistics are not fixed. They change from one sample to another because every sample is different. Statistics are the estimates we use to approximate the hidden population parameters.

- A statistic is a descriptive measure of a sample.
- Statistics vary from sample to sample, depending on which units are selected.

Example: If we select 200 students from a school and calculate their average height, the average ($\bar{x}$) is a statistic.

- Since we can calculate statistics directly from sample data, they are known values.
- Statistics help us to estimate parameters of the population.

• Examples of Statistic include:

1. Sample Mean $\rightarrow \bar{x}$
2. Sample Standard Deviation $\rightarrow s$
3. Sample Variance $\rightarrow s^2$
4. Sample Proportion $\rightarrow \hat{p}$ (p-hat)
5. Sample Median $\rightarrow m$

Example 1 Average Marks of Students

A university has 50,000 students. Instead of calculating marks of all students, suppose we take a random sample of 500 students and compute their average marks. That average = sample mean ($\bar{x}$).

- Population = 50,000 students
- Sample = 500 students
- Statistic ($\bar{x}$) = average marks of the 500 students
- This statistic ($\bar{x}$) is used to estimate the population mean μ .

Example 2 Proportion of Voters Supporting a Candidate

In a state election, there are 3 crore voters. A survey agency takes a random sample of 5,000 voters, out of which 2,100 support Candidate A. So, the sample proportion ($\hat{p}$) = $2100 \div 5000 = 0.42$ or 42%.

- Population = 3 crore voters
- Sample = 5,000 voters
- Sample Proportion ($\hat{p}$) = 42% supporting Candidate A
- This statistic is used to estimate the population proportion P .

Example 3 Defective Rate in a Factory

A factory produces 2,00,000 phones in a month. Instead of checking all, the quality team selects a sample of 1,000 phones. Suppose 15 are found defective. So, the sample defective rate ($\hat{p}$) = $15 \div 1000 = 0.015$ or 1.5%.

- Population = all 2,00,000 phones
- Sample = 1,000 phones
- Sample Proportion Statistic ($\hat{p}$) = defective rate = 1.5%
- This statistic is used to estimate the true defective rate P in the population.



Important Points to Know

- **Parameters** are like the true values we want to know, but usually cannot measure directly because the population is too large.
Parameter = true, fixed value of the population (usually unknown).
- **Statistics** are like the estimates we compute from samples to approximate the parameters.
Statistic = calculated value from a sample (known, but varies).
- **Inferential statistics** is the branch of statistics that helps us draw conclusions about population parameters using the values of sample statistics.
- Since a sample is easier and cheaper to study, we use its results to make educated guesses about the entire population.
- **Inferential Statistics** = Using sample statistics to estimate population parameters

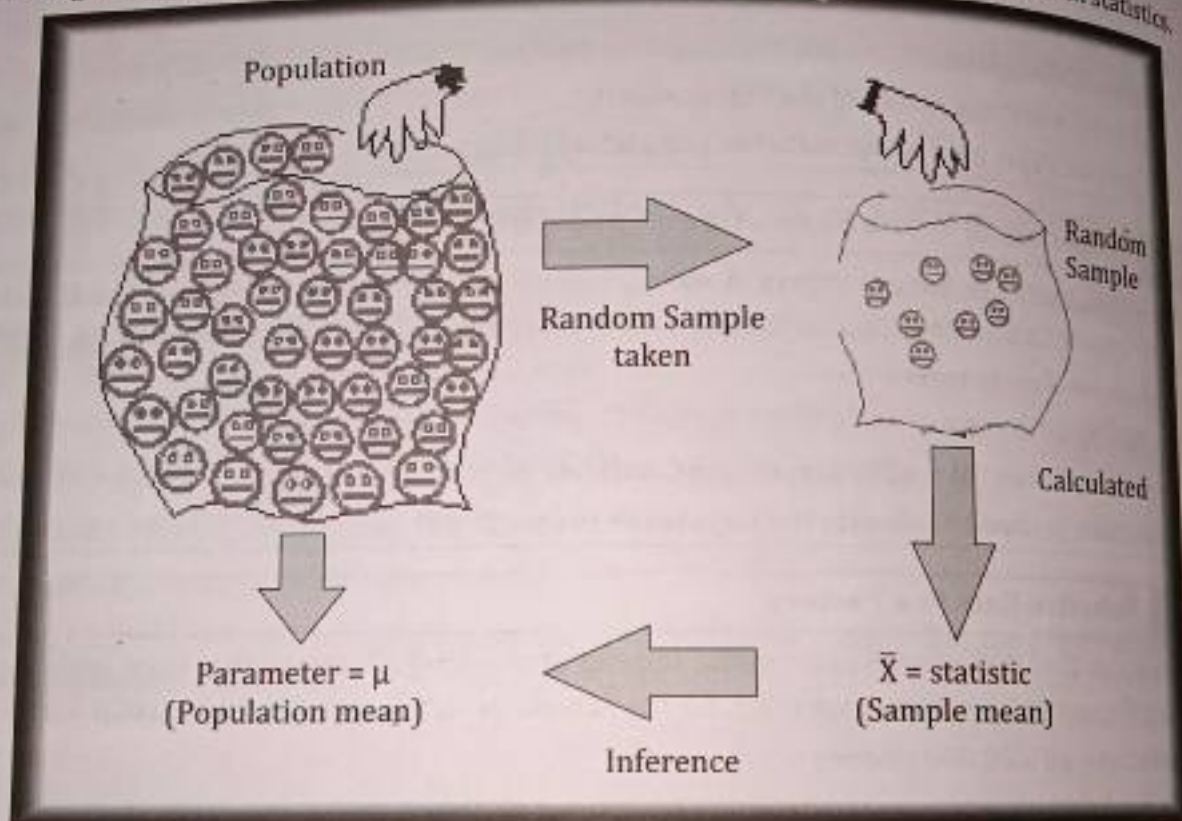
• **Example :**

- Suppose the average height of all 10,000 students in a university is 160 cm. This is the parameter (μ).
- It is not practical to measure all 10,000 students. Instead, we take a sample of 200 students. Their average height is found to be 161 cm. This is the statistic ($\bar{x}$).
- Here, inferential statistics helps us conclude that the population mean μ is approximately 161 cm.



Relationship between a Population, a Sample, and Inference in Statistics.

The below diagram shows the relationship between a population, a sample, and inference in statistics.



1. Population (Left Side)

- The big group shown is the population.
- It contains all individuals, items, or observations of interest.
- The population has a fixed value called a parameter (μ = population mean).
- For example, the average height of all students in a school is μ .

2. Random Sample Taken (Middle Arrow)

- Instead of studying the entire population (which may be too large), a random sample is selected.
- Random sampling means every member of the population has an equal chance of being included.
- **Example:** Selecting 50 students at random from the whole school to measure their heights.

3. Sample (Right Side)

- The smaller group shown is the sample.
- From this sample, we calculate a statistic ($\bar{x}$ = sample mean).
- **Example:** The average height of those 50 students is the statistic $\bar{x}$.

4. Inference

Parameter

Aspect

Meaning

Scope

Value

Known/Unknown

Purpose

Notation

Example

9.5 S

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4. Inference (Bottom Arrow)

- The sample statistic ($\bar{x}$) is then used to make an inference or estimate about the population parameter (μ).
- Since testing the whole population is not possible, inference allows us to draw conclusions about the entire population based on the sample.

Parameter vs Statistic

Aspect	Parameter	Statistic
Meaning	A numerical value that describes a characteristic of the entire population.	A numerical value that describes a characteristic of a sample taken from the population.
Scope	Refers to all units of the population.	Refers to only the selected units in the sample.
Value	Fixed for a population (does not change unless the population changes).	Variable, changes from one sample to another.
Known/Unknown	Usually unknown because studying the entire population is difficult.	Usually known because we calculate it directly from the sample.
Purpose	Represents the true picture of the population.	Provides an estimate of the population parameter.
Notation	Represented by Greek letters ($\mu, \sigma, \sigma^2, P, M$).	Represented by English/other symbols ($\bar{x}, s, s^2, \hat{p}, m$).
Example	Population mean marks of all 50,000 students in a university = μ .	Sample mean marks of 500 students taken from the university = $\bar{x}$.

9.5 Sampling Distributions

In statistics, it is often not possible to study the whole population. Hence, samples are drawn and statistics such as the sample mean, sample proportion, or sample variance are calculated.

When a sample is taken from a population, some statistics are usually calculated such as the sample mean ($\bar{x}$), the sample proportion ($\hat{p}$), or the sample variance (s^2). If another sample of the same size is taken from the same population, the values of these statistics will be different. If this process is continued by taking many samples of the same size and calculating the statistic each time, then all those values of the statistic together will form a distribution. This new distribution is called a **Sampling Distribution**.

- A sampling distribution is always associated with a sample statistic, not with raw observations.
- Different samples taken from the same population will give different statistics.
- The collection of these statistics forms a new distribution called the sampling distribution.
- It tells us how close a sample statistic is likely to be to the population parameter.
- It provides the base for concepts like standard error, confidence intervals, and hypothesis testing.
- It is a fundamental concept for statistical inference, since population parameters are estimated using sample statistics.



Meaning : Sampling Distribution

A Sampling Distribution is the probability distribution of a sample statistic (such as mean, proportion, variance, etc.) obtained by considering all possible samples of a given size drawn from the same population. It shows how a statistic varies from one sample to another and helps in making inferences about the population parameter. In simple words, a sampling distribution describes the behavior of a sample statistic.

Example 1 Average Marks of Students

Suppose a university has 20,000 students. The average marks of all students (μ) is unknown.

- A sample of 200 students gives an average of 62.
- Another sample of 200 students gives an average of 64.
- Another sample gives an average of 61.

If we continue this process and record the sample means, the distribution of these sample means forms the Sampling Distribution of the Mean.

Example 2 Proportion of Defective Bulbs

In a factory, the true defective rate of bulbs is 5%.

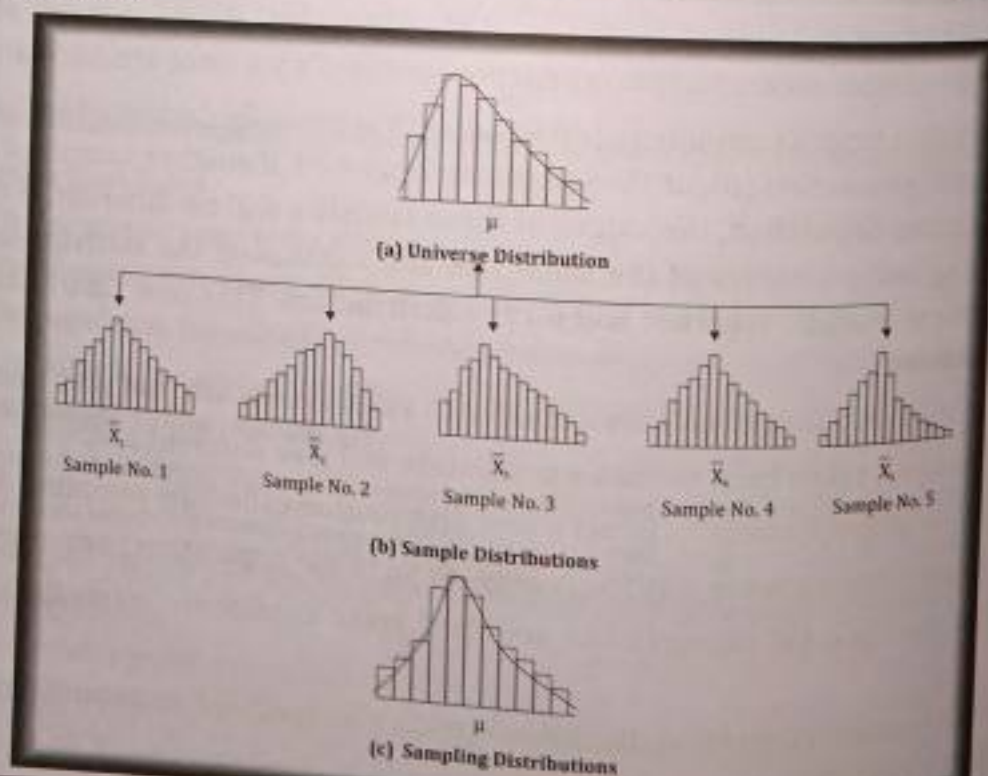
- A sample of 100 bulbs shows 4 defectives (4%).
- Another sample of 100 bulbs shows 7 defectives (7%).
- Another shows 6 defectives (6%).

If we collect these proportions for many samples, they will form the Sampling Distribution of the Proportion.



Understanding Population Distribution, Sample Distribution and Sampling Distribution

The below diagram explains the concept of population distribution (universe distribution), sample distribution and sampling distribution.



(a) Universe Distribution (Population Distribution)

- This is the distribution of the entire population.
- It shows how all individual values in the population are spread around the population mean (μ).
- Universe Distribution = Population Distribution (entire group).
- **Example :** If we are studying the heights of all students in a university, the distribution of heights of all students is the population (or universe) distribution.

(b) Sample Distributions

- From the population, several samples are drawn.
- Each sample has its own distribution and its own mean ($\bar{x}_1, \bar{x}_2, \bar{x}_3, \dots$).
- These sample means may differ from each other because every sample contains different individuals.
- Sample Distributions = Distribution of data within each chosen sample.
- **Example :** If we take 5 samples of 100 students each from the university, each group will have a slightly different average height.
- The diagram shows multiple sample distributions (Sample No. 1, Sample No. 2, Sample No. 3, etc.).

(c) Sampling Distribution

- When we collect all the sample means ($\bar{x}_1, \bar{x}_2, \bar{x}_3, \dots$) from repeated samples and arrange them, they form a new distribution.
- This new distribution is called the Sampling Distribution of the Mean.
- Sampling Distribution = Distribution of statistics (like means or proportions) obtained from many samples.
- It has its own properties, such as:
 - Its mean is equal to the population mean μ .
 - Its spread is measured by the Standard Error (SE), which is smaller than the population's standard deviation.
- **Example :** If the true average height (μ) of all students is 165 cm, then the average of many sample means will also be close to 165 cm.

Example Population Distribution, Sample Distribution and Sampling Distribution

Let us consider a factory defects example to understand the three levels (population distribution, sample distribution, and sampling distribution).

1. Population Distribution (Universe Distribution)

A mobile phone factory produces 20,000 phones in one month.

- Out of these, about 5% are defective.
- If we record the status of all 20,000 phones (defective or not) and plot it, that graph is the Population Distribution.
- It shows the true picture of the whole population.

But in practice, testing all 20,000 phones is very costly and time-consuming.

2. Sample Distributions

Instead of testing all 20,000 phones, the quality control team takes samples.

- Sample 1: Test 200 phones. Found 8 defectives $\rightarrow$ defect rate = 4%.
- Sample 2: Test another 200 phones. Found 14 defectives $\rightarrow$ defect rate = 7%.
- Sample 3: Test another 200 phones. Found 10 defectives $\rightarrow$ defect rate = 5%.
- Sample 4: Test another 200 phones. Found 11 defectives $\rightarrow$ defect rate = 5.5%.

Each sample has its own defect rate (sample proportion $\hat{p}$). These rates are not identical, but they are close to the true population rate of 5%. If we plot each sample separately, we get Sample Distributions.

3. Sampling Distribution

Now suppose we repeat this sampling process hundreds of times, each time selecting 200 phones and calculating the defect proportion.

- Some samples give 4% defectives.
- Some give 5%.
- Some give 6% or 7%.

If we collect all these sample proportions ($\hat{p}$) and plot them together, we get the Sampling Distribution of the Proportion.

The average of all these sample proportions = true population proportion (5%).

The spread of this distribution depends on the sample size. Larger samples give more accurate results with less variation.

Example

Example on Sampling Distribution, Random Error, and Bias

Problem : A college has 4000 students. The true average weight (population mean, μ) of these students is 134.21 lbs. A researcher takes a random sample of 200 students. The average weight of this sample ($\bar{x}$) is 135 lbs. Now, suppose the researcher takes 20 different random samples of 200 students each and calculates their sample means. The means obtained are: 130, 137, 140, 131, 132, 133, 128, 125, 129, 138, 133, 134, 142, 145, 136, 134, 128, 143, 137, 136.

Solution :

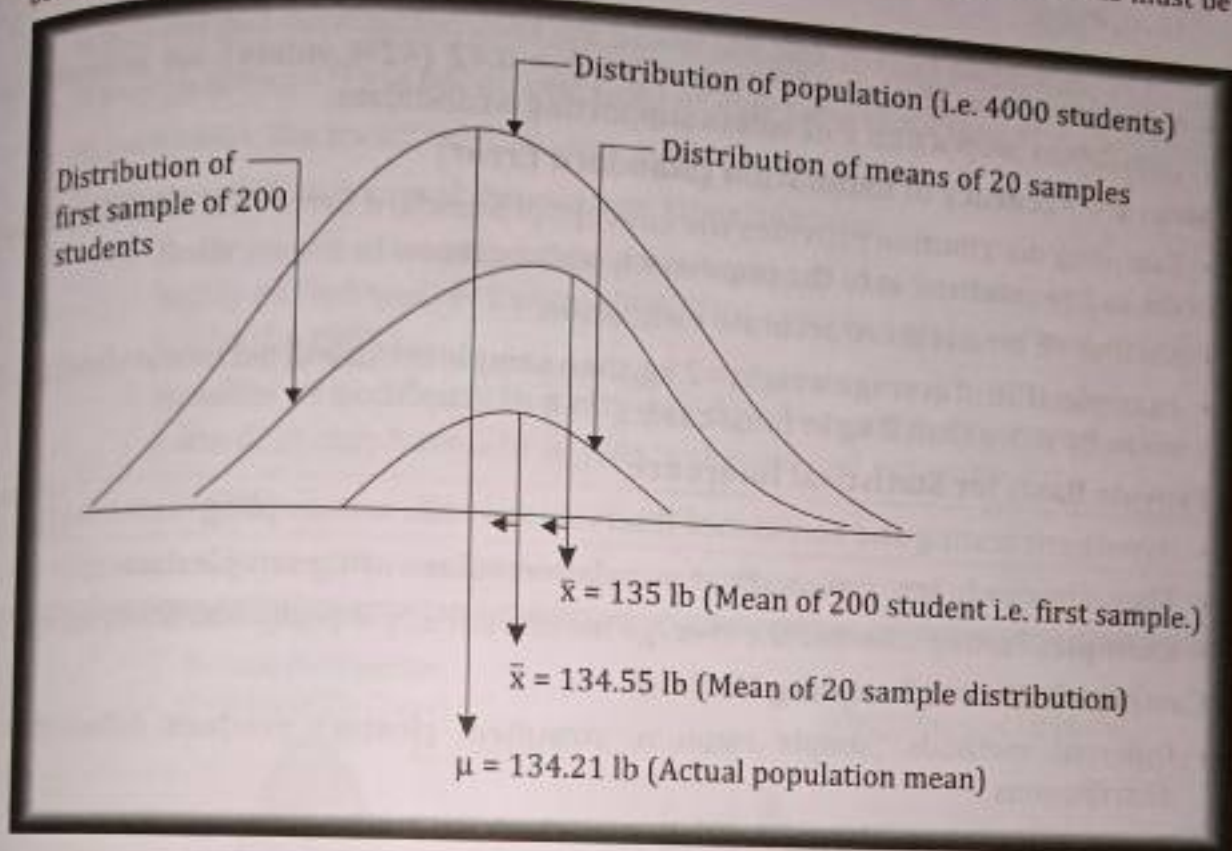
- **Population Mean (Parameter):** The population consists of all 4000 students. Their true average weight is $\mu = 134.21$ lbs.
- **Sample Mean of First Sample (Statistic):** A random sample of 200 students gave mean $\bar{x} = 135$ lbs. This single sample statistic is close to μ but not exact.
- **Sample Means of 20 Samples (Statistic):** When 20 different random samples were taken, each gave a slightly different mean (130, 137, 140, etc.). These means form a sampling distribution of the mean. Their overall average 134.55 lbs is even closer to μ .

Note : One sample mean may not equal the population mean. The mean of many sample means gets closer to the true population mean.

Important Points :

- **Random Error :** The differences between the sample means (like 137, 125, 145) and the true mean $\mu = 134.21$ lbs are due to random error. These errors occur naturally because every random sample is different. Random errors are natural and cancel out with more samples.

- **Systematic Error (Bias)** : If the researcher chooses only healthy students, the sample mean will always be higher than μ . This is not random but a consistent error called **bias**. Bias must be avoided because it gives consistently wrong results.



The diagram shows how the process of sampling distribution works. The curve at the top represents the population distribution of 4000 students with the true mean weight $\mu = 134.21$ lbs. From this population, one random sample of 200 students was taken, and its mean weight was $\bar{x} = 135$ lbs, shown as the distribution of the first sample. If instead of one, 20 different random samples of 200 students each are selected, then their means form a new distribution called the sampling distribution of the sample mean, with mean = 134.55 lbs. This value is even closer to the true population mean compared to the first sample mean. The diagram also highlights that each sample mean may differ from μ due to random error, but as more samples are taken, the mean of sample means gets closer to μ .

9.5.1 Objectives of Sampling Distribution

A sampling distribution is the probability distribution of a statistic (like mean, proportion, variance) obtained from all possible samples of the same size from a population. The objective of sampling distribution is to connect sample statistics with population parameters, measure their reliability, and provide a foundation for inference in statistics.

The main objectives are:

1. To Understand the Behavior of Sample Statistics

- Every sample gives a different value of the statistic (like $\bar{x}$ or $\hat{p}$).
- Sampling distribution helps in studying how these values are spread around the true population parameter.
- **Example:** If many samples of 50 students are taken to calculate average marks, the sampling distribution of $\bar{x}$ will show how these averages vary.

2. To Estimate Population Parameters

- The most important objective is to use sample statistics to estimate population parameters (μ , σ , P).
- It provides a scientific way to generalize from a sample to the population.
- **Example:** From the sample proportion $\hat{p} = 0.42$ (42% voters), we estimate the true population proportion P of voters supporting a candidate.

3. To Measure Accuracy of Estimation (Standard Error)

- Sampling distribution provides the concept of Standard Error (SE) which tells how close the sample statistic is to the population parameter.
- Smaller SE means more accurate estimation.
- **Example:** If SE of average weight = 2 kg, then sample mean will not deviate from population mean by more than 2 kg in most cases.

4. To Provide Basis for Statistical Inference

- Hypothesis testing and confidence intervals are built on sampling distributions.
- They allow us to test claims about population values using sample data.
- **Example:** Testing whether the average life of a battery is really 500 hours using a sample.

5. To Compare Different Sampling Methods

- Different methods (simple random, stratified, cluster) produce different sampling distributions.
- By comparing them, the most efficient method can be chosen.
- **Example:** Stratified sampling often gives a narrower distribution (less SE) than simple random sampling.

6. To Handle Uncertainty in Decision-Making

- Since we cannot study entire populations, sampling distribution provides a way to measure uncertainty.
- It shows the probability of obtaining certain sample results by chance.
- **Example:** If a sample mean is very far from μ in the sampling distribution, it indicates an unusual or rare event.

9.5.2

Normal vs Non-Normal Population Distributions

1. Normal Distribution

- A normal distribution is a symmetrical, bell-shaped curve.
- Most of the values are close to the average (mean), and fewer values lie at the extremes.
- The mean, median, and mode are all the same in a perfectly normal distribution.

• Examples of Normal Population Distributions

- **Human height:** In a large group of students, most heights are around the average (say 165 cm), very few are extremely short or extremely tall.
- **Exam scores:** In a well-designed test, most students score around the average with fewer students getting very high or very low marks.

- **Measurement errors:** Small experimental errors in physics or chemistry usually follow a normal curve.

2. Non-Normal Distribution

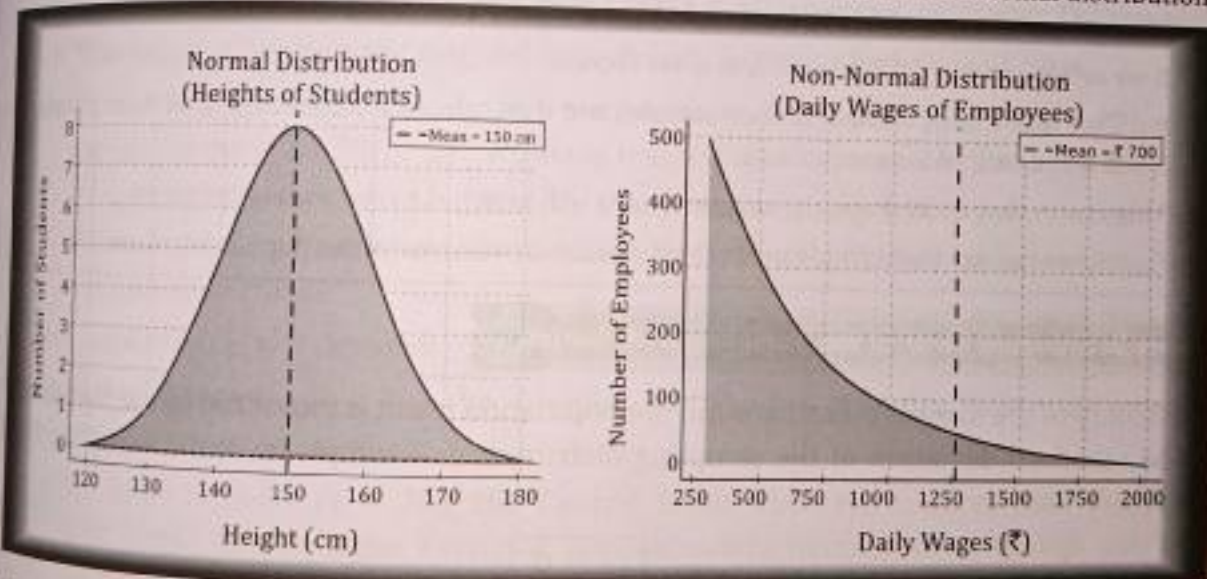
- A non-normal distribution does not follow the bell-shaped pattern.
- It may be skewed (long tail on one side) or may have more than one peak (bimodal).
- In such cases, the mean, median, and mode are not equal.

• Examples of Non-Normal Population Distributions

- **Daily wages of workers:** Many workers may earn between 400 and 600, but a few highly skilled workers earn 2000. This creates a right-skewed distribution (long tail on the right).
- **Number of accidents in a city per day:** Many days may have 0 or 1 accident, but a few days may have 5 or 6. This is skewed and not symmetric.

Example Normal vs Non-Normal Population Distributions

The below diagrams explain the difference between a normal distribution and a non-normal distribution.



1. Normal Distribution (Left - Heights Example)

- The curve is symmetrical and bell-shaped.
- The mean (150 cm) is exactly at the center (red dashed line).
- Most students have heights close to 150 cm, and very few are extremely short or tall.
- In this type of distribution, mean = median = mode.

2. Non-Normal Distribution (Right - Wages Example)

- The curve is right-skewed.
- Most employees earn between ₹400-₹600, while a few highly skilled ones earn very high wages (₹1500-₹2000).
- The mean (~₹700) is pulled to the right (dashed line), away from the peak.
- In skewed distributions, the mean, median, and mode are not equal.

9.5.3

Properties of Sampling Distribution

The concept of sampling distribution is very important because it explains how sample statistics behave when repeated samples are drawn from the same population. Its main properties are:

Property 1: Mean of Sampling Distribution

The mean of the sampling distribution of sample means ($\mu_{\bar{x}}$) is equal to the mean of the population (μ).

$$\mu_{\bar{x}} = \mu$$

This means that if we keep drawing samples of the same size from a population and calculate the mean of each sample, then the average of all those sample means will be exactly equal to the population mean.

Example Property 1: Mean of Sampling Distribution

Suppose the average height of all 10,000 students in a college is 165 cm (this is the population mean μ).

- If we select a sample of 50 students and calculate the average height, it may be 163 cm or 167 cm.
- If we select another 50 students, it may be 166 cm.
- But if we take a large number of such samples and then calculate the average of all those sample means, it will be exactly 165 cm, equal to μ .
- This shows that on average, the sample means will match the true population mean.
- Hence, we can say that sample mean is an unbiased estimator of the population mean.

Property 2: Standard Error of the Sampling Distribution

The variability of the sample means around the population mean is measured by the **Standard Error (SE)**. The standard deviation of the sampling distribution of sample means is called the **Standard Error (SE)**.

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Where:

- σ = population standard deviation
- n = sample size

Standard deviation is a measure of how much individual items in the population differ from the population mean. For example, if $\sigma = 20$ g, it means that the weights of items (say apples or packets) typically vary 20 g above or below the average weight.

Example Property 2: Standard Error of the Sampling Distribution

Suppose the weights of 5,000 packets of sugar have:

- Population mean (μ) = 500 g
- Population standard deviation (σ) = 20 g

If we take random samples:

- For sample size $n = 25$ packets:

$$\sigma_{\bar{x}} = \frac{20}{\sqrt{25}} = \frac{20}{5} = 4\text{g}$$

This means that the average weight of 25 packets will usually vary about ± 4 g from the true mean.

- For sample size $n = 100$ packets:

$$\sigma_{\bar{x}} = \frac{20}{\sqrt{100}} = \frac{20}{10} = 2\text{g}$$

Now the sample mean will usually vary only about ± 2 g from the true mean.
Larger samples give more stable averages with less variability.

Property 3: Shape of the Sampling Distribution

1. If the population is normal:

- The sampling distribution of the sample mean is also normal, no matter the sample size.
- This happens because the original data already follows a bell-shaped curve.
- **Example 1:** If student exam scores are normally distributed ($\mu = 60$, $\sigma = 10$), then the sample means of 20, 50, or 100 students will all follow a normal bell-shaped distribution.
- **Example 2:** The heights of students in a large class usually follow a normal distribution. If we take repeated samples of 30 or 40 students, the distribution of their average heights will also be normal.

2. If the population is not normal:

- For small samples ($n < 30$), the sampling distribution may also look non-normal, with skewness or irregular shape.
- For large samples ($n \geq 30$), the Central Limit Theorem ensures that the sampling distribution of the mean becomes approximately normal, even though the original population is not.
- **Example 1:** Wages of workers may be skewed, but if we take samples of 50 or 100 workers, the average wages (sample means) will be close to a bell-shaped distribution.
- **Example 2:** The number of daily accidents in a city is often skewed (many days have 0 or 1 accident, but a few days have 6 or 7). If we take large samples of 40 or more days, the distribution of sample means of accidents becomes nearly normal.

Property 4: Effect of Sample Size

The larger the sample size (n), the smaller the variability of the sample means, and the more reliable the estimate of the population mean.

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Thus, the accuracy of the sample mean depends on:

- The sample size (n) $\rightarrow$ larger n reduces SE.
- The variability in the population (σ) $\rightarrow$ more spread in the population gives higher SE.

Example Property 4: Effect of Sample Size

Suppose the average height of students is $\mu = 165$ cm, $\sigma = 10$ cm.

- If $n = 25$ students:

$$\sigma_{\bar{x}} = \frac{10}{\sqrt{25}} = 2 \text{ cm}$$

- If $n = 100$ students:

$$\sigma_{\bar{x}} = \frac{10}{\sqrt{100}} = 1 \text{ cm}$$

With $n = 25$, sample means may vary by ± 2 cm.

With $n = 100$, sample means may vary only by ± 1 cm.

So larger samples give more precision in estimating the population mean.

9.6 Sampling Distribution of Sample Mean

When we take different random samples from the same population, each sample will generally have a different mean ($\bar{x}$) because the elements included in one sample may not be the same as in another.

This means:

- A population has one fixed mean (μ). But many different samples can be drawn from it, and each will have a different mean ($\bar{x}$). The distribution of these sample means is the sampling distribution of $\bar{x}$.
- The sample mean ($\bar{x}$) is not fixed. It is a random variable, because it depends on the random choice of units in the sample. Since $\bar{x}$ is a random variable, it has its own probability Distribution, which is known as the Sampling Distribution of the Sample Mean.
- In general, the sampling distribution of sample means depending on the distribution of the population or process from which samples are drawn.

Note: Just like the mean, other statistics such as the median, mode, proportion, or standard deviation also have their own sampling distributions.

Example 1: Sampling Distribution of Sample Mean

Let us take an example, suppose there are 5 workers A, B, C, D and E comprising the population. Their monthly wages in thousand rupees are:

A	B	C	D	E
3	5	7	7	8

$$\text{Mean of the population} = \frac{3+5+7+7+8}{5} = \frac{30}{5} = 6$$

Now, we select samples of size $n = 3$. The possible combinations and their means are:

Sample	Wages in the Sample	Sample Mean ($\bar{X}$)
ABC	3, 5, 7	5.00
ABD	3, 5, 7	5.00
ABE	3, 5, 8	5.33
ACD	3, 7, 7	5.67
ACE	3, 7, 8	6.00
ADE	3, 7, 8	6.00
BCD	5, 7, 7	6.33
BCE	5, 7, 8	6.67
BDE	5, 7, 8	6.67
CDE	7, 7, 8	7.33

Sample Distribution of Means : Now group the sample means as shown below.

Mean	5.0	5.33	5.67	6.0	6.33	6.67	7.33
Frequency	2	1	1	2	1	2	1

Mean of sampling distribution of means :

$$= \frac{5 \times 2 + 5.33 \times 1 + 5.67 \times 1 + 6 \times 2 + 6.33 \times 1 + 6.67 \times 2 + 7.33 \times 1}{2 + 1 + 1 + 2 + 1 + 2 + 1}$$

$$= \frac{10 + 5.33 + 5.67 + 12 + 6.33 + 13.34 + 7.33}{10}$$

$$= \frac{60}{10} = 6 = \text{Mean of the population}$$

Note : The mean of the population (μ) and the mean of the sampling distribution of the mean ($\mu_{\bar{x}}$) is the same but the SD of population and the SD of the sampling distribution of the mean are not the same.

Example 2 : Students' Test Scores

Consider 4 students with marks: 40, 50, 60, 70.

Population mean (μ) = $(40 + 50 + 60 + 70) \div 4 = 55$.

If we take samples of size $n = 2$ (without replacement):

Sample	Scores	Sample Mean ($\bar{x}$)
(40, 50)	40, 50	45
(40, 60)	40, 60	50
(40, 70)	40, 70	55
(50, 60)	50, 60	55
(50, 70)	50, 70	60
(60, 70)	60, 70	65

Sampling distribution of means: {45, 50, 55, 55, 60, 65}.

- Mean of sample means = $(45 + 50 + 55 + 55 + 60 + 65) \div 6 = 55$.
- This matches the population mean $\mu = 55$.

This shows that the sample mean is an unbiased estimator of the population mean.

Example 3: Light Bulb Lifetimes

Suppose a factory produces 3 light bulbs with lifetimes (in hours): {1000, 1200, 1400}.

Population mean (μ) = $(1000 + 1200 + 1400) \div 3 = 1200$.

If we take samples of size $n = 2$:

Sample	Lifetimes	Sample Mean ($\bar{x}$)
(1000, 1200)	1000, 1200	1100
(1000, 1400)	1000, 1400	1200
(1200, 1400)	1200, 1400	1300

Sampling distribution of means = {1100, 1200, 1300}.

- Mean of these sample means = $(1100 + 1200 + 1300) \div 3 = 1200$.
- Again, this equals the population mean μ .

Illustration 1

The ages (in years) of three employees in a small shop are : {25, 35, 45}

Take all possible samples of size 2 with replacement. Construct the sampling distribution of the sample mean and verify that the mean of the sampling distribution equals the population mean.

Solution

Step 1 : Find the Population mean

Population values : 25, 35, 45

$$\mu = \frac{25 + 35 + 45}{3} = \frac{105}{3} = 35$$

So, population mean = 35 years

Step 2 : List all possible samples of size 2 (with replacement)

Since each draw has 3 choices and we draw 2 times, total samples = $3^2 = 9$.

Sample	Values	Sample Mean
1	(25, 25)	25
2	(25, 35)	30
3	(25, 45)	35
4	(35, 25)	30
5	(35, 35)	35

7	(45, 25)	35
8	(45, 35)	40
9	(45, 45)	45

Step 3: Construct frequency distribution of sample means

Sample mean $\bar{X}$	Frequency	Probability
25	1	1/9
30	2	2/9
35	3	3/9
40	2	2/9
45	1	1/9

Step 4: Find Expected Value of Sample Mean

$$\begin{aligned}
 E[\bar{X}] &= \sum \bar{x} \cdot P(\bar{x}) \\
 &= 25 \cdot \frac{1}{9} + 30 \cdot \frac{2}{9} + 35 \cdot \frac{3}{9} + 40 \cdot \frac{2}{9} + 45 \cdot \frac{1}{9} \\
 &= \frac{25 + 60 + 105 + 80 + 45}{9} = \frac{315}{9} = 35
 \end{aligned}$$

$$\text{So, } E[\bar{X}] = 35 = \mu$$

This proves that the sampling distribution mean equals the population mean.

Illustration 2

Population values : {10, 20, 30}.

Sampling is with replacement.

Sample size $n = 3$.

1. Construct the sampling distribution of the sample mean.
2. Find the $E[\bar{X}]$ and verify it equals the population mean.

Solution

Step 1: Find the population mean

$$\mu = \frac{10 + 20 + 30}{3} = 20$$

Population mean equals 20.

Step 2 : List all samples of size 3 and their means

Total number of ordered samples = $3^3 = 27$.

Case	Samples	Sample mean $\bar{x}$	Frequency
All same (all 10)	(10,10,10)	10	1
All same (all 20)	(20,20,20)	20	1
All same (all 30)	(30,30,30)	30	1
Two 10s and one 20	(10,10,20)(10,20,10)(20,10,10)	$40/3 \approx 13.33$	3
Two 10s and one 30	(10,10,30)(10,30,10)(30,10,10)	$50/3 \approx 16.67$	3
Two 20s and one 10	(20,20,10)(20,10,20)(10,20,20)	$50/3 \approx 16.67$	3
Two 20s and one 30	(20,20,30)(20,30,20)(30,20,20)	$70/3 \approx 23.33$	3
Two 30s and one 10	(30,30,10)(30,10,30)(10,30,30)	$70/3 \approx 23.33$	3
Two 30s and one 20	(30,30,20)(30,20,30)(20,30,30)	$80/3 \approx 26.67$	3
All different	All 6 permutations of (10,20,30)	20	6

Total frequency = $1 + 1 + 1 + 3 + 3 + 3 + 3 + 3 + 3 + 6 = 27$.

Step 3 : Build the sampling distribution of sample means

Sample mean $\bar{x}$	Frequency	Probability $P(\bar{x})$
10	1	$1/27$
$40/3$	3	$3/27$
$50/3$	6	$6/27$
20	7	$7/27$
$70/3$	6	$6/27$
$80/3$	3	$3/27$
30	1	$1/27$
Total	27	1

Step 4 : Compute the mean of the sampling distribution

$$E[\bar{X}] = \sum \bar{x} P(\bar{x})$$

$\bar{x}$	$P(\bar{x})$	$\bar{x} \times P(\bar{x})$
10	$1/27$	$10/27$
$40/3$	$3/27$	$40/27$
$50/3$	$6/27$	$100/27$
20	$7/27$	$140/27$
$70/3$	$6/27$	$140/27$
$80/3$	$3/27$	$80/27$
30	$1/27$	$30/27$
Sum	1	$540/27 = 20$

So $E[\bar{X}] = 20$.

This equals the population mean and hence the sample mean is an unbiased estimator.

Illustration 3

Population values are {2, 4, 6, 8}.

Sample size is $n=2$.

Sampling is without replacement.

1. Construct the sampling distribution table of the sample mean.
2. Find the mean of this distribution. Verify that it equals the population mean.

Solution

Step 1. Population mean

$$\mu = \frac{2+4+6+8}{4} = 5$$

Population mean equals 5.

Step 2. List all samples of size 2 without replacement and their means

Total unordered samples equals

Sample	Values	Sample mean $\bar{x}$
1	(2, 4)	3
2	(2, 6)	4
3	(2, 8)	5
4	(4, 6)	5
5	(4, 8)	6
6	(6, 8)	7

Step 3. Sampling distribution of sample means

Each sample has probability $1/6$

Sample mean $\bar{x}$	Frequency	Probability $P(\bar{x})$
3	1	$1/6$
4	1	$1/6$
5	2	$2/6$
6	1	$1/6$
7	1	$1/6$
Total	6	1

Step 4. Mean of the sampling distribution

$$\begin{aligned} E[\bar{X}] &= \sum \bar{x}P(\bar{x}) = 3 \cdot \frac{1}{6} + 4 \cdot \frac{1}{6} + 5 \cdot \frac{2}{6} + 6 \cdot \frac{1}{6} + 7 \cdot \frac{1}{6} \\ &= \frac{3+4+10+6+7}{6} = \frac{30}{6} = 5 \end{aligned}$$

So, $E[\bar{X}] = 5$.

Population mean and hence the sample mean is unbiased

Illustration 4

Population values are {4 6 8 10 12}

Sample size is $n=3$.

Sampling is without replacement.

1. Construct the sampling distribution table of the sample mean.
2. Find the mean of this distribution. Verify that it equals the population mean.

Solution

Step 1. Population mean

$$\mu = \frac{4+6+8+10+12}{5} = \frac{40}{5} = 8$$

Population mean equals 8.

Step 2. List all samples of size 3 without replacement and their means

Total unordered samples equals $\binom{5}{3} = 10$

Sample	Values	Sample Mean $\bar{x}$
1	(4 6 8)	6
2	(4 6 10)	20/3
3	(4 6 12)	22/3
4	(4 8 10)	22/3
5	(4 8 12)	8
6	(4 10 12)	26/3
7	(6 8 10)	8
8	(6 8 12)	26/3
9	(6 10 12)	28/3
10	(8 10 12)	10

Total samples equal 10.

Step 3. Sampling distribution of sample means

Each sample has probability $1/10$.

Sample mean	Frequency	Probability
6	1	1/10
20/3	1	1/10
22/3	2	2/10
8	2	2/10
26/3	2	2/10
28/3	1	1/10
10	1	1/10
Total	10	1

Step 4. Mean of the sampling distribution

$$E[\bar{X}] = \sum \bar{x}P(\bar{x}) = \frac{1}{10} \left(6 + \frac{20}{3} + 2 \cdot \frac{22}{3} + 2 \cdot 8 + 2 \cdot \frac{26}{3} + \frac{28}{3} + 10 \right)$$

$$= \frac{1}{10} \cdot \frac{1}{3} (18 + 20 + 44 + 48 + 52 + 28 + 30) = \frac{1}{10} \cdot \frac{240}{3} = \frac{80}{10} = 8$$

$$\text{So } E[\bar{X}] = 8$$

It equals the population mean and hence the sample mean is unbiased.

Illustration 5

Population values are {3, 5, 7, 9}

Sample size is $n = 3$

Sampling is without replacement

1. Construct the sampling distribution table of the sample mean.
2. Find the mean of this distribution. Verify that it equals the population mean.

Solution**Step 1: Population mean**

$$\mu = \frac{3+5+7+9}{4} = \frac{24}{4} = 6$$

Population mean equals 6.

Step 2: List all samples of size 3 without replacement and their means

Total unordered samples equal $\binom{4}{3} = 4$.

Sample	Values	Sample mean $\bar{x}$
1	(3 5 7)	5
2	(3 5 9)	17/3
3	(3 7 9)	19/3
4	(5 7 9)	7

Step 3: Sampling distribution of sample means

Each sample has probability 1/4.

Sample mean $\bar{x}$	Frequency	Probability $P(\bar{x})$
5	1	1/4
17/3	1	1/4
19/3	1	1/4
7	1	1/4
Total	4	1

Step 4: Mean of the Sampling Distribution

$$E[\bar{X}] = \sum \bar{x}P(\bar{x}) = \frac{1}{4} \left(5 + \frac{17}{3} + \frac{19}{3} + 7 \right) = \frac{1}{4} \cdot \frac{1}{3} (15 + 17 + 19 + 21) = \frac{1}{4} \cdot \frac{72}{3} = \frac{24}{4} = 6$$

So $E[\bar{X}] = 6$

It equals the population mean and hence the sample mean is unbiased.

9.6.1 Sampling Distribution of Mean When Population has Non-normal Distribution

In statistics, populations are not always normally distributed. For example,

- The daily wages of workers may be right-skewed, because most workers earn between ₹400 and ₹600, while a few skilled workers earn more than ₹2000.
- The number of accidents per day in a city may follow a Poisson pattern, where most days have 0, 1, or 2 accidents, but a few days have 5 or 6.
- The family expenditure on luxury goods may be highly uneven, since most families spend very little while a few families spend large amounts.

In such cases, the shape of the population distribution is not a bell curve. It may be skewed, irregular, or multi-peaked.

Now, when we take samples from such a non-normal population, the values of the sample mean ($\bar{x}$) will also vary. If the sample size is small, the distribution of these sample means will look similar to the population — skewed or irregular. But as the sample size grows larger, the **Central Limit Theorem (CLT)** helps us. It states that the sampling distribution of the mean becomes approximately normal, even when the population itself is not normal.

9.6.1.1 Central Limit Theorem (CLT)

The Central Limit Theorem is one of the most important results in statistics. It explains why the normal distribution is so widely used in real-world data analysis. The theorem states that:

**Central Limit Theorem**

The theorem states that:

- When the sample size is sufficiently large, the sampling distribution of the sample mean ($\bar{x}$) can be approximated by a normal distribution, regardless of the shape of the original population distribution.
- In other words, even if the population is skewed, irregular, or non-normal, the distribution of sample means will approach a bell-shaped curve when enough observations are taken.

Formal Statement :

- The central limit theorem states that the re-averaged sum of a sufficiently large number of identically distributed independent random variables, each with finite mean and variance, will be approximately normally distributed.

This means that:

- Suppose a population has a finite mean μ and standard deviation σ .

- If we take random samples of size 'n' from this population and compute their sample means ($\bar{x}$), then as 'n' increases, the sampling distribution of sample mean $\bar{X}$ is approximately normally distributed, with mean and standard deviation as:

$$\mu_{\bar{x}} = \mu \text{ and } \sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

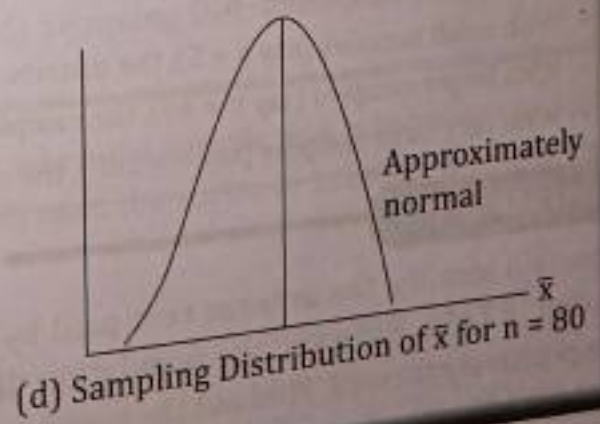
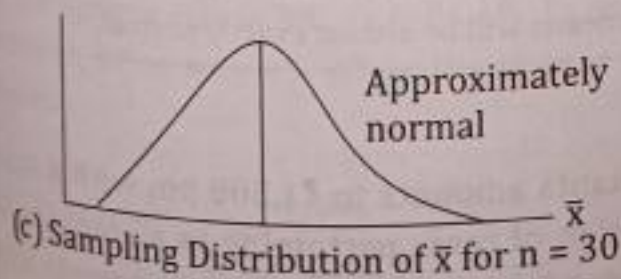
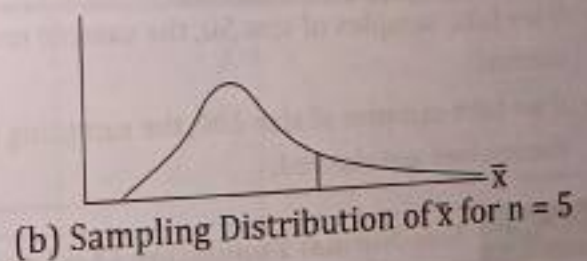
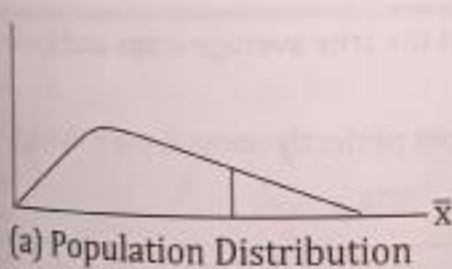
The theorem plays a crucial role in statistical inference, because it allows us to draw conclusions about population parameters using random samples, even when the population itself is not normally distributed.

A natural question arises: What sample size is considered large enough for the Central Limit Theorem to hold? In general, a sample size of 30 or more observations is usually taken as sufficient for applying the Central Limit Theorem in most practical situations. However, the exact minimum sample size depends on how close the population is to normal:

- If the population distribution is highly skewed or very different from normal, a larger sample size is required to achieve a good normal approximation.
- If the population distribution is already close to normal, even a smaller sample size may be enough.

To conclude, the larger the sample size, the better the approximation of the sampling distribution to the normal curve.

The below figure shows a number of situations with varying sample sizes. It shows the shape of sampling distributions with different sample sizes. It is clear from the figure that as the sample size increases, the shape of the sampling distribution undergoes change and it tends to be more and more approximate to normal distribution.



(a) Population Distribution

- The population is not normal.
- It may be skewed or irregular.
- Mean = μ .

(b) Sampling Distribution for $n = 5$

- Samples of size 5 are taken.
- The sample mean distribution is still skewed, not symmetric.
- Small samples do not remove the irregularities of the population.

(c) Sampling Distribution for $n = 30$

- Samples of size 30 are taken.
- The sample means now form an approximately normal distribution.
- The skewness has almost disappeared.

(d) Sampling Distribution for $n = 80$

- Samples of size 80 are taken.
- The distribution of sample means is now almost perfectly normal.
- It is centered at μ , and the variability is very small.
- This shows that with large samples, the sample mean is a very reliable estimate of the population mean.

Example 1: Skewed Population (Daily Wages)

Suppose the daily wages of workers in a factory follow a right-skewed distribution. Most earn around ₹500, but a few earn more than ₹2000.

- If we take samples of size 5, the sample means are still skewed (because one high wage can affect the mean a lot).
- If we take samples of size 50, the sample means cluster around the true average wage and look nearly normal.
- If we take samples of size 200, the sampling distribution is almost perfectly normal, even though wages themselves are skewed.

Example 2: Non-Normal Population (Accidents per Day)

Suppose the number of accidents per day in a city follows a Poisson distribution. Most days have 0–2 accidents, but a few days have 5–6.

- With small samples (say $n = 5$), the distribution of sample means will still look uneven.
- With larger samples (say $n = 40$), the sample means will form an approximately normal curve.
- With very large samples (say $n = 100$), the sample means will be almost exactly normal.

Illustration 1

In a certain locality, the average rent paid by all tenants amounts to ₹1,500 pm with a standard deviation of ₹450. However, the population distribution of rents pertaining to all tenants in that city is positively skewed. Find out the mean and standard deviation of $\bar{X}$ when the sample size is (i) 30 and (ii) 100. Also describe the shape of its sampling distribution in both cases.

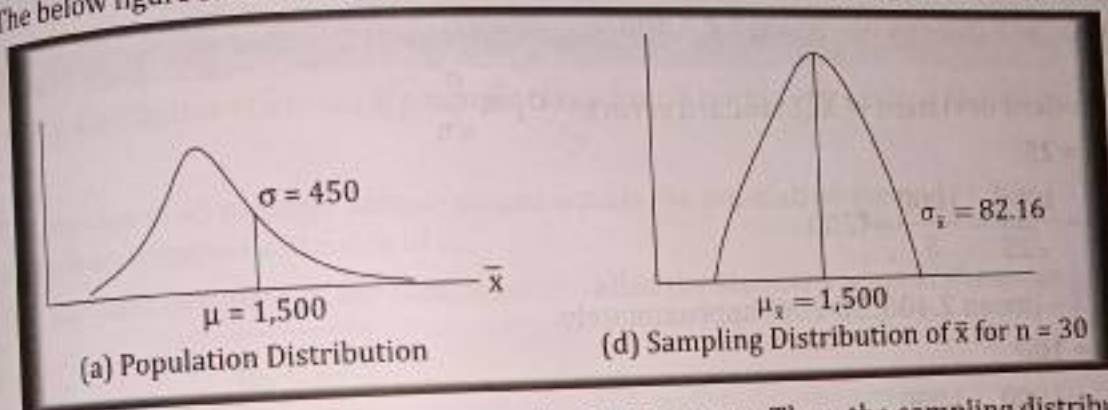
Solution

Given that the population distribution is not normal, but the sample size in both the cases is large as $n \geq 30$, the central limit theorem can be applied to infer the shape of the sampling distribution of $\bar{X}$.

- (i) Let $\bar{X}$ be the average rent paid by a sample of 30 tenants. Then, the sampling distribution of $\bar{X}$ is approximately normal with the values of the mean and standard deviation as a

$$\mu_{\bar{X}} = \mu = ₹1,500 \text{ and } \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{450}{\sqrt{30}} = 82.16$$

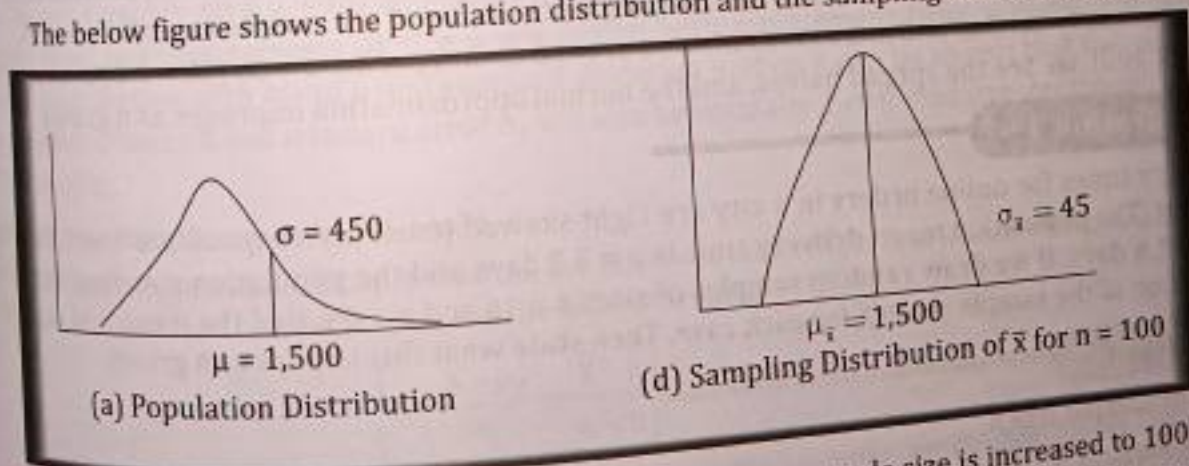
The below figure shows the population distribution and the sampling distribution of $\bar{X}$.



- (ii) Let $\bar{X}$ be the average rent paid by a sample of 100 tenants. Then, the sampling distribution of $\bar{X}$ is approximately normal with the values of the mean and standard deviation as

$$\mu_{\bar{X}} = \mu = \text{Rs } 1500 \text{ and } \sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{450}{\sqrt{100}} = 45$$

The below figure shows the population distribution and the sampling distribution of $\bar{X}$.



When the sample size is 30, $\sigma_{\bar{X}} = 82.16$. In contrast, when the sample size is increased to 100, $\sigma_{\bar{X}}$ is much lower, being only 45. When the sample size increase, the variance between $\bar{X}$ and its mean μ decreases.

Illustration 2

In a large city, individual monthly electricity bills are right-skewed (many small bills, a few very large ones). The population mean bill is ₹ 2,400 with population standard deviation ₹ 1,000. If we draw random samples of sizes $n = 25$ and $n = 100$, find the mean and standard deviation of the sample mean $\bar{X}$ for each case. Then state what this implies about the sampling distribution as n grows.

Solution

For the sample mean $\bar{X}$:

- Mean of $\bar{X}$ (for any n): $\mu_{\bar{X}} = \mu = ₹ 2,400$

- Standard deviation of $\bar{X}$ (Standard error): $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$

Case 1: $n = 25$

$$\sigma_{\bar{X}} = \frac{1000}{\sqrt{25}} = \frac{1000}{5} = ₹ 200$$

So, $\bar{X} \sim (\text{mean } 2,400, \text{SD } 200)$ approximately.

Case 2: $n = 100$

$$\sigma_{\bar{X}} = \frac{1000}{\sqrt{100}} = \frac{1000}{10} = ₹ 100$$

So, $\bar{X} \sim (\text{mean } 2,400, \text{SD } 100)$ approximately.

Although the population of bills is non-normal (right-skewed), the Central Limit Theorem tells us that as the sample size increases, the sampling distribution of the sample mean $\bar{X}$ becomes approximately normal and more concentrated around μ . Comparing $n = 25$ ($SE = ₹ 200$) to $n = 100$ ($SE = ₹ 100$), we see the spread halves, and the normal approximation improves as n grows.

Illustration 3

Delivery times for online orders in a city are right-skewed (most arrive quickly, a few take much longer). The population mean delivery time is $\mu = 3.2$ days and the population standard deviation is $\sigma = 1.6$ days. If we draw random samples of sizes $n = 16$ and $n = 64$, find the mean and standard deviation of the sample mean $\bar{X}$ for each case. Then state what this implies as n grows.

Solution

For any sample size n ,

- Mean of $\bar{X}$: $\mu_{\bar{X}} = \mu = 3.2$ days

- Standard deviation (standard error) of $\bar{X}$: $\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$

Case 1: $n = 16$

$$\sigma_{\bar{X}} = \frac{1.6}{\sqrt{16}} = \frac{1.6}{4} = 0.4 \text{ days}$$

So, $\bar{X}$ has mean 3.2 days and SD 0.4 days (approximately).

Case 2: $n = 64$

$$\sigma_{\bar{x}} = \frac{1.6}{\sqrt{64}} = \frac{1.6}{8} = 0.2 \text{ days}$$

So, $\bar{X}$ has mean 3.2 days and SD 0.2 days (approximately).

Even though individual delivery times are non-normal (right-skewed), the Central Limit Theorem implies that the sampling distribution of $\bar{X}$ becomes approximately normal as 'n' increases, and it becomes tighter around μ . Going from $n = 16$ to $n = 64$, the standard error halves ($0.4 \rightarrow 0.2$). This shows the improved normal approximation and reduced variability of the sample mean.

9.6.2 Sampling Distribution of Mean When Population has Normal Distribution

The sampling distribution of a mean of a sample taken from a normal population shows two important properties.

- The average of all possible sample means equals the population mean. ($\mu_{\bar{x}} = \mu$). The sample mean is an unbiased estimator of μ .
- The standard deviation of the sample mean called the standard error of the mean is

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$$

Where,

$\sigma_{\bar{x}}$ = standard error of the mean

σ = population standard deviation

n = sample size

The larger samples give a smaller standard error, so sample means vary less from sample to sample. If all possible samples of size n are drawn with replacement from a population having normal distribution with mean μ and standard deviation σ , then it can be shown that the sampling distribution of mean $\bar{X}$ and standard error $\sigma_{\bar{x}}$ will also be normally distributed irrespective of the size of the sample.

Converting a sample mean to a Z-score

- To find how unusual a sample mean is, the observed mean ($\bar{X}$) is first standardized to a Z-score on the standard normal scale using the below formula:

$$Z = \frac{\bar{X} - \mu_{\bar{x}}}{\sigma_{\bar{x}}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

- Here, z indicates how many standard errors the sample mean lies above (positive z) or below (negative z) the population mean. A larger $|z|$ means a more extreme mean relative to μ . (Z tells how unusual $\bar{X}$ is under the assumed μ and σ .)

Making inferences with $\bar{X}$ when σ is known

- The procedure for making statistical inference using sampling distribution about the population mean μ based on mean $\bar{X}$ of sample means is summarized as follows:
- If the population standard deviation σ value is known and either
 1. population distribution is normal, or

- ii. Population is not normal but the sample size is large (rule of thumb $n \geq 30$; by the Central Limit Theorem the distribution of $\bar{X}$ is then close to normal)

In both cases, use

$$Z = \frac{\bar{X} - \mu_{\bar{X}}}{\sigma_{\bar{X}}} = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

Finite Population Correction (when sampling without replacement)

- If the population is finite with N elements whose mean is μ and variance is σ^2 and the samples of fixed size ' n ' are drawn without replacement, then the standard deviation (also called standard error) of sampling distribution of mean $\bar{X}$ can be modified to adjust the continued change in the size of the population N due to the several draws of samples of size n as follows:

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

The term $\sqrt{(N-n)/(N-1)}$ is called the finite population correction factor (fpc).



Note:

This factor has little effect on reducing the amount of sampling error when the size of the sample is less than 5 percent of the population size. But if N is large relative to the sample size n , $\sqrt{(N-n)/(N-1)}$ is approximately equal to 1.

Illustration 1

A factory packs 1 kg rice packets. The weight of each packet follows a Normal distribution with:

- Mean $\mu = 1000$ grams
- Standard deviation $\sigma = 20$ grams

A sample of size $n = 36$ is taken.

1. What is the mean and standard deviation of the sampling distribution of sample mean?
2. Find the probability that sample mean is between 995 and 1005 grams.
3. Find the probability that sample mean is greater than 1004 grams.

Solution

Step 1: Given Values : $\mu = 1000$; $\sigma = 20$; $n = 36$

Step 2: Mean and standard Error of sample mean

Formula:

$$\mu_{\bar{X}} = \mu$$

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

Substitute:

$$\mu_{\bar{x}} = 1000$$

$$\sigma_{\bar{x}} = \frac{20}{\sqrt{36}} = \frac{20}{6} = 3.33$$

So:

- Mean of sampling distribution = 1000 g
- Standard deviation (Standard Error) = 3.33 g

Step 3: Probability that $995 \leq \bar{X} \leq 1005$

Formula for Z-score:

$$Z = \frac{\bar{X} - \mu_{\bar{x}}}{\sigma_{\bar{x}}}$$

For $\bar{X} = 995$:

$$Z_1 = \frac{995 - 1000}{3.33} = \frac{-5}{3.33} = -1.5$$

For $\bar{X} = 1005$:

$$Z_2 = \frac{1005 - 1000}{3.33} = \frac{5}{3.33} = 1.5$$

Use Normal table (area from 0 to z)

From the Z-table:

- Area between 0 and 1.5 = 0.4332

$$\text{So, } P(Z \leq 1.5) = 0.5 + 0.4332 = 0.9332$$

$$P(Z \leq -1.5) = 0.5 - 0.4332 = 0.0668$$

Required Probability:

$$\begin{aligned} P(995 \leq \bar{X} \leq 1005) &= P(Z \leq 1.5) - P(Z \leq -1.5) \\ &= 0.9332 - 0.0668 = 0.8664 \end{aligned}$$

Probability = 0.8664 or 86.64%

Step 4: Probability that $\bar{X} \geq 1004$

For $\bar{X} = 1004$

$$Z = \frac{1004 - 1000}{3.33} = \frac{4}{3.33} = 1.2$$

From the Z-table:

- Area between 0 and 1.2 = 0.3849

So:

$$P(Z \leq 1.2) = 0.5 + 0.3849 = 0.8849$$

$$P(Z \geq 1.2) = 1 - 0.8849 = 0.1151$$

Probability = 0.1151 or 11.51%

Final Answer:

1. $\mu_{\bar{X}} = 1000, \sigma_{\bar{X}} = 3.33$
2. $P(995 \leq \bar{X} \leq 1005) = 0.8664$ or 86.64%
3. $P(\bar{X} \geq 1004) = 0.1151$ or 11.51%

**Note:**

The standard normal curve is symmetric with 0.5 area on each side of 0.

- For a positive z :

Area to the left of z = (left half up to 0) + (area from 0 to z)

$$P(Z \leq z) = 0.5 + A(z) \quad (z > 0).$$

This is a "plus" because we add the small slice from 0 to z to the 0.5 on the left.

- For a negative z :

Area to the left of z = (left half up to 0) - (area from 0 to $|z|$)

$$P(Z \leq z) = 0.5 - A(|z|) \quad (z < 0).$$

This is a "minus" because the left endpoint is to the left of 0, so we subtract the slice from 0 back to $|z|$.

Now put these two together for an interval (a, b) that crosses 0 with $a < 0 < b$:

$$P(a < Z < b) = P(Z \leq b) - P(Z \leq a) = [0.5 + A(b)] - [0.5 - A(|a|)] = A(b) + A(|a|).$$

Illustration 2

In a normally distributed population the average income per household is Rs 10000 per month. The population standard deviation is Rs 800. A random sample of size $n=100$ households is taken. Find the probability that the sample mean income lies between Rs 9800 and Rs 10100.

Solution:**Step 1: Given values**

$$\mu = 10000 \quad \sigma = 800 \quad n = 100$$

Step 2: Mean and Standard Error of the sample mean

$$\mu_{\bar{X}} = \mu = 10000$$

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{800}{\sqrt{100}} = \frac{800}{10} = 80$$

So the sampling distribution of $\bar{X}$ is Normal with mean 10000 and standard deviation 80.

Step 3: Convert bounds to Z scores

General formula

$$Z = \frac{\bar{X} - \mu_{\bar{X}}}{\sigma_{\bar{X}}}$$

Lower bound Rs. 9800

$$Z_1 = \frac{9800 - 10000}{80} = \frac{-200}{80} = -2.5$$

Upper bound Rs. 10100

$$Z_2 = \frac{10100 - 10000}{80} = \frac{100}{80} = 1.25$$

Step 4: Use the Normal table with the +0.5 and -0.5 adjustments
Most standard Normal tables give area from 0 to z.

Area from 0 to 1.25 is 0.3944.

Hence

$$P(Z \leq 1.25) = 0.5 + 0.3944 = 0.8944$$

Area from 0 to 2.5 is 0.4938.

Hence

$$P(Z \leq -2.5) = 0.5 - 0.4938 = 0.0062$$

Step 5: Required probability

$$\begin{aligned} P(9800 \leq \bar{X} \leq 10100) &= P(Z \leq 1.25) - P(Z \leq -2.5) \\ &= 0.8944 - 0.0062 = 0.8882 \end{aligned}$$

So the probability is 0.8882. That is 88.82%

This is the probability that the sample mean will lie between Rs 9,800 and Rs 10,100. This is shown in the below figure.

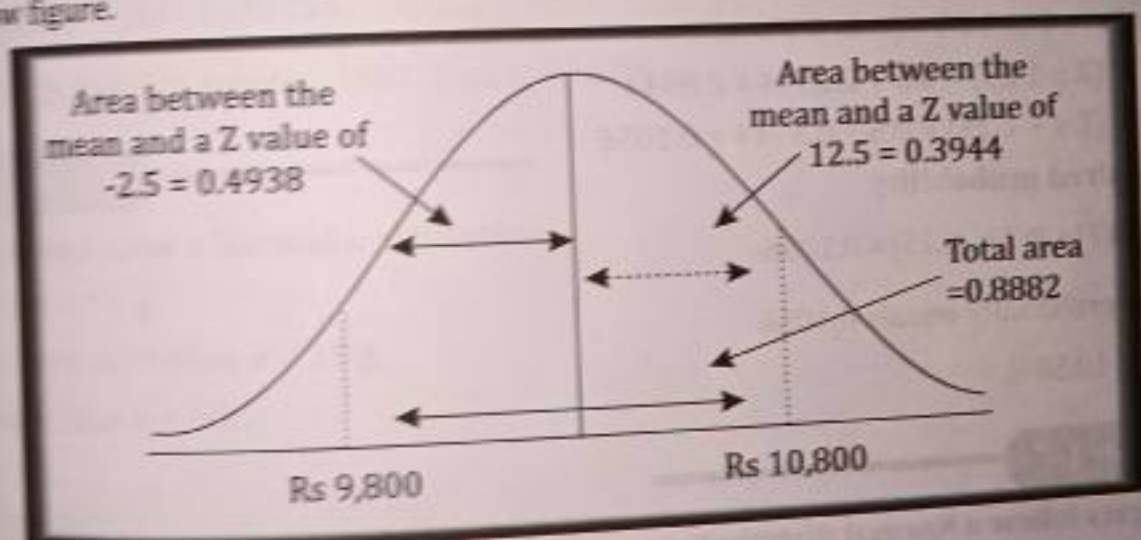


Illustration 3

A machine fills packets. Packet weights are normally distributed.

Mean $\mu = 50$ g.

Known standard deviation $\sigma = 12$ g.

Sample size $n = 25$.

Find $P(\bar{X} < 47)$.

Solution

Step 1 : Given values : $\mu = 50, \sigma = 12, n = 25$

Step 2 : Mean and Standard Error of $\bar{X}$

$$\mu_{\bar{X}} = \mu = 50$$

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}} = \frac{12}{\sqrt{25}} = \frac{12}{5} = 2.4$$

So the sampling distribution of $\bar{X}$ is normal with mean 50 and standard deviation 2.4.

Step 3 : Convert the bound to a Z score

General formula

$$Z = \frac{\bar{X} - \mu_{\bar{X}}}{\sigma_{\bar{X}}}$$

For $\bar{X} = 47$

$$Z = \frac{47 - 50}{2.4} = \frac{-3}{2.4} = -1.25$$

Step 4 : Use the Normal table with +0.5 and -0.5

Most standard Normal tables give the area from 0 to z.

Area from 0 to 1.25 equals 0.3944.

$$P(Z \leq 1.25) = 0.5 + 0.3944 = 0.8944$$

$$P(Z \leq -1.25) = 0.5 - 0.3944 = 0.1056$$

Step 5 : Required probability

$$P(\bar{X} < 47) = P(Z < -1.25) = 0.1056$$

So the probability equals 0.1056.

That is 10.56%

Illustration 4

Packet weights follow a Normal distribution.

- Mean $\mu = 200$ g.
- Standard deviation $\sigma = 20$ g.
- Sample size $n = 16$.

Find $P(\bar{X} > 205)$.

Solution

Step 1 : Given values : $\mu = 200, \sigma = 20, n = 16$

Step 2 : Mean and Standard Error of $\bar{X}$

$$\mu_{\bar{X}} = \mu = 200$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{20}{\sqrt{16}} = \frac{20}{4} = 5$$

So the sampling distribution of $\bar{X}$ is Normal with mean 200 and standard deviation 5.

Step 3: Convert the bound to a Z score

$$\text{General formula: } Z = \frac{\bar{X} - \mu_{\bar{x}}}{\sigma_{\bar{x}}}$$

$$\text{For } \bar{X} = 205, Z = \frac{205 - 200}{5} = \frac{5}{5} = 1$$

Step 4: Use the Normal table with +0.5 and -0.5

Most Standard Normal tables give the area from 0 to z.

Area from 0 to 1.0 equals 0.3413

$$P(Z \leq 1.0) = 0.5 + 0.3413 = 0.8413$$

$$P(Z \geq 1.0) = 1 - 0.8413 = 0.1587$$

Step 5: Required probability

$$P(\bar{X} > 205) = P(Z > 1.0) = 0.1587$$

So the probability equals 0.1587. That is 15.87%

Illustration 5

Packet weights follow a Normal distribution.

- Mean $\mu = 75$ g.
- Standard deviation $\sigma = 15$ g.
- Sample size $n = 36$.

Find $P(\bar{X} < 72)$.

Solution

Step 1: Given Values: $\mu = 75, \sigma = 15, n = 36$

Step 2: Mean and Standard Error of $\bar{X}$

$$\mu_{\bar{x}} = \mu = 75$$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{15}{\sqrt{36}} = \frac{15}{6} = 2.5$$

The Sampling distribution of $\bar{X}$ is Normal. Mean equals 75 and Standard deviation equals 2.5

Step 3: Convert the bound to a Z score

$$\text{General Formula: } Z = \frac{\bar{X} - \mu_{\bar{x}}}{\sigma_{\bar{x}}}$$

$$\text{For } \bar{X}=72, \quad Z = \frac{72-75}{2.5} = \frac{-3}{2.5} = -1.2$$

Step 4 : Use the Normal table with + 0.5 and - 0.5

Most standard Normal tables give the area from 0 to z.

Area from 0 to 1.2 equals 0.3849.

$$P(Z \leq 1.2) = 0.5 + 0.3849 = 0.8849$$

$$P(Z \leq -1.2) = 0.5 - 0.3849 = 0.1151$$

Step 5 : Required probability

$$P(\bar{X} < 72) = P(Z < -1.2) = 0.1151$$

So the probability equals 0.1151. That is 11.51%

Illustration 6

A corporation tracks staff turnover across 20 branches of equal size.

The population standard deviation of annual turnover is 40 employees.

A sample of 4 branches is selected without replacement.

Find the standard error of the mean.

Solution

Step 1. Write the formula

Standard error of the mean with finite population correction : $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$

Step 2. Substitute the values

$$\sigma_{\bar{x}} = \frac{40}{\sqrt{4}} \sqrt{\frac{20-4}{20-1}} = \frac{40}{2} \sqrt{\frac{16}{19}}$$

Step 3. Simplify

$$\sigma_{\bar{x}} = 20 \sqrt{\frac{16}{19}}$$

Step 4. Numerical value

$$\sqrt{\frac{16}{19}} = \sqrt{0.842105} \approx 0.917663$$

$$\sigma_{\bar{x}} \approx 20 \times 0.917663 \approx 18.35 \text{ employees}$$

Standard error of the mean ≈ 18.35 employees.

Illustration 7

Suppose we are interested in 15 electronic companies of the same size. All these companies are confronting a serious problem in the form of excessive turnover of their employees. It has been found that the standard deviation of the distribution of annual turnover is 60 employees. Take a sample of 3 electronic companies, without replacement, and determine the standard error of the mean.

Solution

Given:

Population size $N=15$.

Population standard deviation $\sigma=60$ employees.

Sample size $n=3$.

Sampling is without replacement.

Standard error of the mean with finite population correction :

$$\begin{aligned}\sigma_{\bar{x}} &= \frac{\sigma}{\sqrt{n}} \times \sqrt{\frac{N-n}{N-1}} = \frac{60}{\sqrt{3}} \times \sqrt{\frac{15-3}{15-1}} \\ &= \frac{60}{1.7321} \times \sqrt{0.8571} \\ &= \frac{60}{1.7321} \times 0.9258 \\ &= 34.64 \times 0.9258 = 32.07\end{aligned}$$

Thus, the standard error is 32.07 employees.

Illustration 8

The mean length of life of a certain cutting tool is 41.5 hours with a standard deviation of 2.5 hours. What is the probability that a simple random sample of size 50 drawn from this population will have a mean between 40.5 hours and 42 hours?

Solution

Step 1. Standard error of the mean

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{2.5}{\sqrt{50}} = \frac{2.5}{7.0711} \approx 0.3536$$

The sampling distribution of $\bar{X}$ is Normal with mean 41.5 and standard deviation 0.3536

Step 2. Convert bounds to Z with substitution

$$\text{General formula : } Z = \frac{\bar{X} - \mu_{\bar{X}}}{\sigma_{\bar{X}}}$$

$$\text{Lower bound } 40.5 : Z_1 = \frac{40.5 - 41.5}{0.3536} = \frac{-1.0}{0.3536} \approx -2.83$$

$$\text{Upper bound } 42.0 : Z_2 = \frac{42.0 - 41.5}{0.3536} = \frac{0.5}{0.3536} \approx 1.41$$

Step 3. Use Normal table with the + 0.5 and -0.5 adjustments

Most standard tables give area from 0 to z.

For $z = 1.41$, Area from 0 to 1.41 is 0.4207 is $P(Z \leq 1.41) = 0.5 + 0.4207 = 0.9207$

For $z = 2.83$, Area from 0 to 2.83 is 0.4977 is $P(Z \leq -2.83) = 0.5 - 0.4977 = 0.0023$

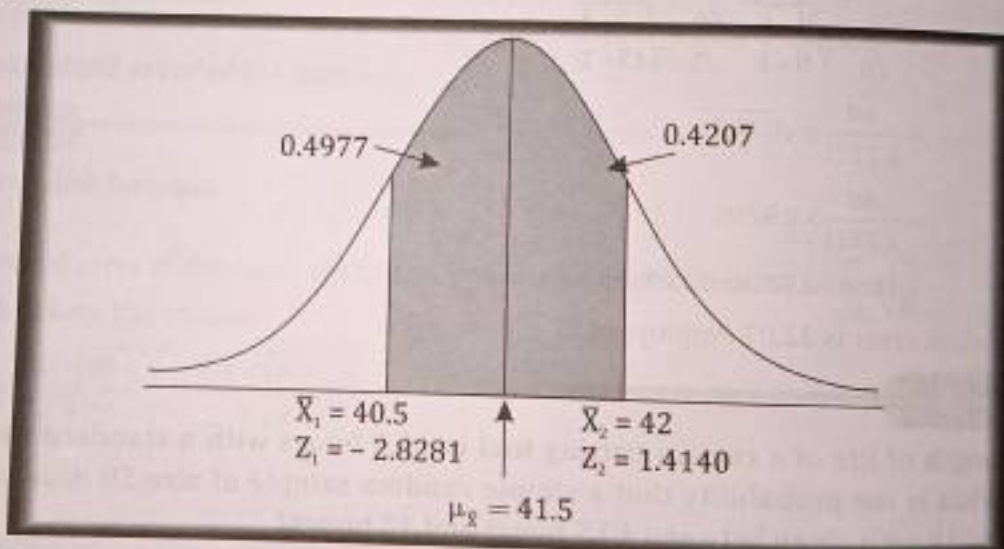
Step 4. Required probability

$$P(40.5 \leq \bar{X} \leq 42) = P(Z \leq 1.41) - P(Z \leq -2.83) = 0.9207 - 0.0023 = 0.9184$$

Probability ≈ 0.9184

That is 91.84%.

The normal distribution can be used to find the required probability as shown by the shaded area in the below figure



9.6.3 Sampling Distribution of Difference Between two Sample Means

The concept of sampling distribution of sample mean introduced earlier in this chapter can also be used to compare a population of size N_1 having mean μ_1 and standard deviation σ_1 with another similar type of population of size N_2 having mean μ_2 and standard deviation σ_2 .

Let $\bar{X}_1$ and $\bar{X}_2$ be the mean of sampling distribution of mean of two populations respectively. Then the difference between their mean values μ_1 and μ_2 can be estimated by generalizing the formula of standard normal variable as follows:

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{X}_1 - \bar{X}_2}} = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{X}_1 - \bar{X}_2}}$$

where,

- $\mu_{\bar{x}_1 - \bar{x}_2} = \mu_{\bar{x}_1} - \mu_{\bar{x}_2} = \mu_1 - \mu_2$ (mean of sampling distribution of difference of two means)

- $\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\sigma_{\bar{x}_1}^2 + \sigma_{\bar{x}_2}^2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$ (Standard error of sampling distribution of two means)

- n_1, n_2 = independent random samples drawn from first and second population, respectively.

Since random samples are drawn independently from two populations with replacement, therefore the sampling distribution of the difference of two means $\bar{x}_1 - \bar{x}_2$ will be normal provided sample size of sufficiently large.

Illustration 1

Car stereos of manufacturer A have a mean lifetime of 1400 hours with a standard deviation of 200 hours, while those of manufacturer B have a mean lifetime of 1200 hours with a standard deviation of 100 hours. If a random sample of 125 stereos of each manufacturer are tested, what is the probability that manufacturer A's stereos will have a mean lifetime which is at least 160 hours than manufacturer B's stereos?

Solution

We are given the following information

Manufacturer A: $\mu_1 = 1400$ hours, $\sigma_1 = 200$ hours, and $n_1 = 125$

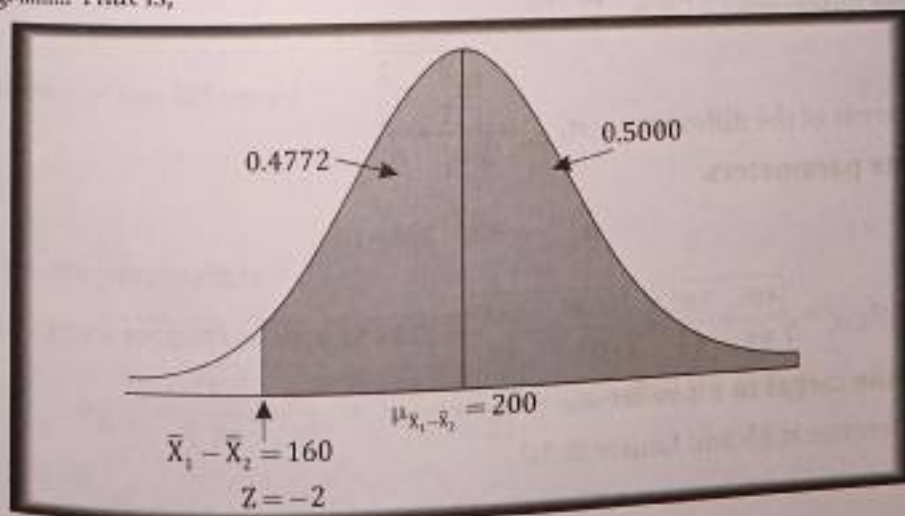
Manufacturer B: $\mu_2 = 1200$ hours, $\sigma_2 = 100$ hours, and $n_2 = 125$

Thus $\mu_{\bar{x}_1 - \bar{x}_2} = \mu_{\bar{x}_1} - \mu_{\bar{x}_2} = \mu_1 - \mu_2 = 1400 - 1200 = 200$

$$\sigma_{\bar{x}_1 - \bar{x}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} = \sqrt{\frac{(200)^2}{125} + \frac{(100)^2}{125}} = \sqrt{80 + 320} = \sqrt{400} = 20$$

Let $\bar{x}_1 - \bar{x}_2$ be the difference in mean lifetime of stereo manufactured by the two manufacturers.

Then we are required to find the probability that this difference is more than or equal to 160 hours, as shown in fig. That is,



$$\begin{aligned}
 P\left[(\bar{X}_1 - \bar{X}_2) \geq 160\right] &= P\left[z \geq \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sigma_{\bar{X}_1 - \bar{X}_2}}\right] \\
 &= P\left[z \geq \frac{160 - 200}{20}\right] \\
 &= P[z \geq -2] \\
 &= 0.5000 + 0.4772 = 0.9772 \text{ (Area under normal curve)}
 \end{aligned}$$

Hence, the probability is very high that mean lifetime of the stereos of A is 160 hours more than that of B.

Illustration 2

Two smartphone brands are tested for battery life.

Population 1. Mean $\mu_1 = 300$ hours. Standard deviation $\sigma_1 = 40$ hours.

Population 2. Mean $\mu_2 = 290$ hours. Standard deviation $\sigma_2 = 50$ hours.

Independent random samples are taken.

Sample sizes are $n_1 = 64$ and $n_2 = 49$.

Find $P((\bar{X}_1 - \bar{X}_2) > 15)$.

Solution

Step 1. Formulas

$$z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Mean of the difference : $\mu_{\bar{X}_1 - \bar{X}_2} = \mu_1 - \mu_2$

Standard error of the difference : $\sigma_{\bar{X}_1 - \bar{X}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$

Step 2. Substitute parameters

$$\mu_{\bar{X}_1 - \bar{X}_2} = 300 - 290 = 10$$

$$\sigma_{\bar{X}_1 - \bar{X}_2} = \sqrt{\frac{40^2}{64} + \frac{50^2}{49}} = \sqrt{\frac{1600}{64} + \frac{2500}{49}} = \sqrt{25 + 51.0204} = \sqrt{76.0204} \approx 8.718$$

Step 3. Convert the target to a z value

Target difference is 15 and Center at 10.

$$z = \frac{15 - 10}{8.718} = \frac{5}{8.718} \approx 0.573$$

Step 4. Use the Normal table with +0.5 and -0.5

Standard tables give area from 0 to z.

For $z=0.57$ the area from 0 to z is about 0.2157.

$$P(Z \leq 0.57) = 0.5 + 0.2157 = 0.7157$$

Right tail

$$P(Z \leq 0.57) = 1 - 0.7157 = 0.2843$$

Step 5. Conclusion

$$P((\bar{X}_1 - \bar{X}_2) > 15) \approx 0.2843$$

So the probability is about 28.43%.

Illustration 3

Two independent populations are considered.

Population 1. $\mu_1=82$, $\sigma_1=12$. Sample size $n_1=36$.

Population 2. $\mu_2=78$, $\sigma_2=15$. Sample size $n_2=25$.

Find $P(2 \leq \bar{X}_1 - \bar{X}_2 \leq 8)$ Use the same formula as in the note.

Solution

Step 1. Write the required formula

$$z = \frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Mean of the difference

$$\mu_{\bar{X}_1 - \bar{X}_2} = \mu_1 - \mu_2$$

Standard error of the difference

$$\sigma_{\bar{X}_1 - \bar{X}_2} = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

Step 2. Substitute the parameters

$$\begin{aligned} \mu_{\bar{X}_1 - \bar{X}_2} &= 82 - 78 = 4 \\ \sigma_{\bar{X}_1 - \bar{X}_2} &= \sqrt{\frac{12^2}{36} + \frac{15^2}{25}} = \sqrt{\frac{144}{36} + \frac{225}{25}} = \sqrt{4 + 9} = \sqrt{13} \approx 3.606 \end{aligned}$$

Step 3. Convert the bounds to zLower bound $a=2$,

$$z_1 = \frac{2-4}{3.606} = \frac{-2}{3.606} = -0.555$$

Upper bound $b=8$

$$z_2 = \frac{8-4}{3.606} = \frac{4}{3.606} = 1.109$$

Step 4. Use the Normal table with +0.5 and -0.5For $z = 1.11$ Area from 0 to 1.11 is 0.3665.

$$P(Z \leq 1.11) = 0.5 + 0.3665 = 0.8665$$

For $z = 0.55$ Area from 0 to 0.55 is 0.2088.

$$P(Z \leq -0.55) = 0.5 - 0.2088 = 0.2912$$

Step 5. Required probability

$$P(2 \leq \bar{X}_1 - \bar{X}_2 \leq 8) = P(Z \leq 1.11) - P(Z \leq -0.55) = 0.8665 - 0.2912 = 0.5753$$

Probability ≈ 0.5753 . That is 57.53%.**9.7 Sampling Distribution of Sample Proportion**

This section extends the idea of sampling distribution from the sample mean to the sample proportion. In previous chapter, we discussed the binomial distribution where p is taken as the success while $q = 1 - p$ is taken as the failure.

Suppose a random sample of size ' n ' is taken from a population. Let ' x ' denote the number of successes in the sample. Then the sample proportion is

$$\bar{p} = \frac{\text{Elements of sample}}{\text{Sample size}} = \frac{x}{n}$$

This statistic is used to estimate the population proportion p .

When the sample size is large ($n \geq 30$), this sampling distribution of the sample proportion can be approximated by a normal distribution.

- **Mean of the sampling distribution :**

$$\mu_{\bar{p}} = p$$

The sample proportion is therefore an unbiased estimator of the population proportion.

- **Standard deviation or standard error (For sampling with replacement or for a very large population)**

$$\sigma_{\bar{p}} = \sqrt{\frac{p(1-p)}{n}} = \frac{pq}{n}$$

• Finite population correction for sampling without replacement

The mean and standard deviation (error) of the sampling distribution of proportion are valid for a finite population in which sampling is with replacement. However, for finite population in which sampling is done without replacement, we have

$$\mu_{\bar{p}} = p \quad \text{and} \quad \sigma_{\bar{p}} = \sqrt{\frac{pq}{n}} \sqrt{\frac{N-n}{N-1}}$$

Where, $q = 1 - p$

• Standardization to the standard normal variable

To standardize sample proportion $\bar{p}$, the standard normal variable:

$$Z = \frac{\bar{p} - \mu_{\bar{p}}}{\sigma_{\bar{p}}} = \frac{\bar{p} - p}{\sqrt{p(1-p)/n}}$$

is approximately the standard normal distribution.

Illustration 1

A manufacturer of watches has determined from experience that 3 per cent of the watches he produces are defective. If a random sample of 300 watches is examined, what is the probability that the proportion defective is between 0.02 and 0.035?

Solution

Step 1 : Parameters

$$p = 0.03, q = 1 - p = 0.97, n = 300$$

Step 2 : Mean and standard error

$$\mu_{\bar{p}} = p = 0.03$$

$$\sigma_{\bar{p}} = \sqrt{\frac{pq}{n}} = \sqrt{\frac{0.03 \times 0.97}{300}} = \sqrt{0.000097} \approx 0.00985$$

Step 3 : Convert bounds to z and round as requested

General formula :

$$z = \frac{\bar{p} - p}{\sigma_{\bar{p}}}$$

Lower bound 0.02

$$z_1 = \frac{0.02 - 0.03}{0.00985} \approx -1.02$$

Upper bound 0.035

$$z_2 = \frac{0.035 - 0.03}{0.00985} \approx 0.51$$

Step 4 : Use Normal table with + 0.5 and - 0.5

Area from 0 to 0.51 is 0.1950

$$P(Z \leq 0.51) = 0.5 + 0.1950 = 0.6950$$

Area from 0 to 1.02 is 0.3461

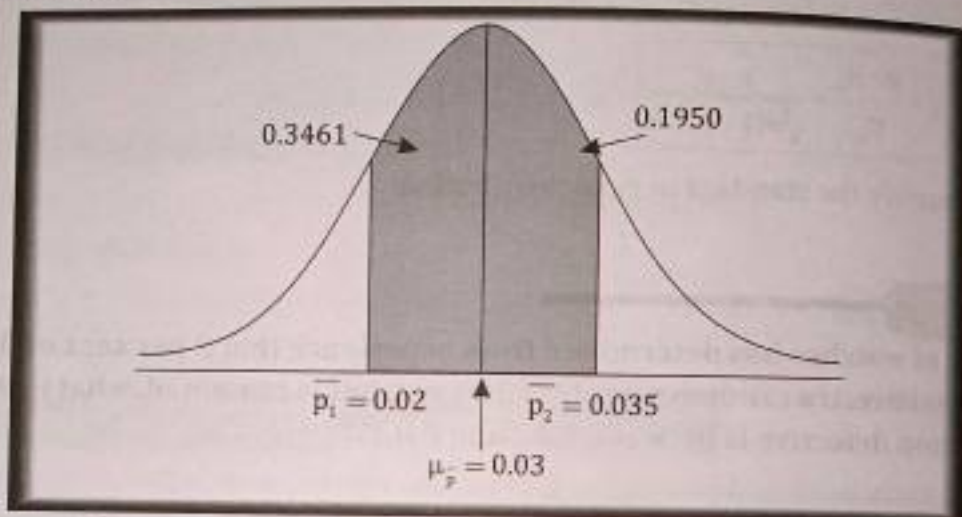
$$P(Z \leq -1.02) = 0.5 - 0.3461 = 0.1539$$

Step 5 : Required probability

$$P(0.02 \leq \bar{p} \leq 0.035) = P(Z \leq 0.51) - P(Z \leq -1.02) = 0.6950 - 0.1539 = 0.5411$$

Probability ≈ 0.5411

That is 54.11%

**Illustration 2**

A Factory produces items with defect rate $p = 0.15$

A random sample of $n = 250$ items is inspected.

Find $P(0.12 \leq \bar{p} \leq 0.18)$

Solution**Step 1 : Parameters**

$$p = 0.15, q = 1 - p = 0.85, n = 250$$

Step 2 : Mean and standard error

$$\mu_{\bar{p}} = p = 0.15$$

$$\sigma_{\bar{p}} = \sqrt{\frac{pq}{n}} = \sqrt{\frac{0.15 \times 0.85}{250}} = \sqrt{0.00051} \approx 0.02258$$

Step 3 : Convert bounds to z

General formulam:

$$z = \frac{\bar{p} - p}{\sigma_{\bar{p}}}$$

Lower bound

$$z_1 = \frac{0.12 - 0.15}{0.02258} = \frac{-0.03}{0.02258} = -1.33$$

Upper bound 0.18

$$z_2 = \frac{0.18 - 0.15}{0.02258} = \frac{0.03}{0.02258} = 1.33$$

Step 4: Normal table with + 0.5 and - 0.5

Area from 0 to 1.33 is 0.4082

$$P(Z \leq 1.33) = 0.5 + 0.4082 = 0.9082$$

$$P(Z \leq -1.33) = 0.5 - 0.4082 = 0.0918$$

Step 5: Required probability

$$P(0.12 \leq \bar{P} \leq 0.18) = P(Z \leq 1.33) - P(Z \leq -1.33) = 0.9082 - 0.0918 = 0.8164$$

Probability ≈ 0.8164

That is 81.64%

Illustration 3

A Factory has a small finite lot of $N = 200$ items.

The population defect proportion is $p = 0.20$

A simple random sample of $n = 50$ items is taken without replacement

Find $P(0.16 \leq \bar{P} \leq 0.24)$

Use the finite population correction.

Solution

Step 1: Parameters

$$p = 0.20 \quad q = 1 - p = 0.80 \quad N = 200 \quad n = 50$$

Step 2: Mean and Standard error with FPC

$$\mu_{\bar{p}} = p = 0.20$$

$$\sigma_{\bar{p}} = \sqrt{\frac{pq}{n}} \sqrt{\frac{N-n}{N-1}} = \sqrt{\frac{0.20 \times 0.80}{50}} \sqrt{\frac{200-50}{200-1}}$$

$$= \sqrt{0.0032} \sqrt{\frac{150}{199}} = 0.05657 \times 0.8685 = 0.04911$$

Step 3: Convert bounds to z

General formula

$$z = \frac{\bar{P} - p}{\sigma_{\bar{p}}}$$

Lower bound 0.16

$$z_1 = \frac{0.16 - 0.20}{0.04911} \approx -0.81$$

Upper bound 0.24

$$z_2 = \frac{0.24 - 0.20}{0.04911} \approx 0.81$$

Step 4 : Normal table with +0.5 and -0.5

Area from 0 to 0.81 is 0.2910

$$P(Z \leq 0.81) = 0.5 + 0.2910 = 0.7910$$

$$P(Z \leq -0.81) = 0.5 - 0.2910 = 0.2090$$

Step 5 : Required probability

$$P(0.16 \leq \bar{P} \leq 0.24) = 0.7910 - 0.2090 = 0.5820$$

Probability ≈ 0.5820 . That is 58.20%

Illustration 4

A Ware house has a finite stock of $N = 400$ items

The true defect proportion is $p = 0.12$

A simple random sample of $n = 80$ items is drawn without replacement.

Find $P(0.10 \leq \bar{P} \leq 0.15)$

Use the finite population correction.

Solution

Step 1 : Parameters

$$p = 0.12, q = 1 - p = 0.88, N = 400, n = 80$$

Step 2 : Mean and Standard error with FPC

$$\mu_{\bar{p}} = p = 0.12$$

$$\begin{aligned} \sigma_{\bar{p}} &= \sqrt{\frac{pq}{n}} \sqrt{\frac{N-n}{N-1}} = \sqrt{\frac{0.12 \times 0.88}{80}} \sqrt{\frac{400-80}{399}} \\ &= \sqrt{0.00132} \sqrt{\frac{320}{399}} = 0.03633 \times 0.89556 \approx 0.03254 \end{aligned}$$

Step 3 : Convert bounds to z

General formula

$$z = \frac{\bar{P} - p}{\sigma_{\bar{p}}}$$

Lower bound 0.10

$$z_1 = \frac{0.10 - 0.12}{0.03254} \approx -0.61$$

Upper bound 0.15

$$z_2 = \frac{0.15 - 0.12}{0.03254} = 0.92$$

Step 4: Normal table using +0.5 and -0.5

Area from 0 to 0.92 is 0.3212

$$P(Z \leq 0.92) = 0.5 + 0.3212 = 0.8212$$

Area from 0 to 0.61 is 0.2291

$$P(Z \leq -0.61) = 0.5 - 0.2291 = 0.2709$$

Step 5: Required probability

$$P(0.10 \leq \bar{P} \leq 0.15) = 0.8212 - 0.2709 = 0.5503$$

Probability ≈ 0.5503 . That is 55.03%

9.7.1

Sampling Distribution of the Difference of two Proportions

Suppose two populations of size N_1 and N_2 are given. For each sample of size n_1 from first population, compute sample proportion $\bar{P}_1$ and standard deviation $\sigma_{\bar{P}_1}$. Similarly for each sample of size n_2 from second population, compute sample proportion $\bar{P}_2$ and standard deviation $\sigma_{\bar{P}_2}$.

For all combinations of these samples from these populations, we can obtain a sampling distribution of the difference $\bar{P}_1 - \bar{P}_2$ of samples proportions. Such a distribution is called sampling distribution of difference of two proportions. The mean and standard deviation of this distribution are given by

$$\mu_{\bar{P}_1 - \bar{P}_2} = \mu_{\bar{P}_1} - \mu_{\bar{P}_2} = p_1 - p_2$$

and

$$\sigma_{\bar{P}_1 - \bar{P}_2} = \sqrt{\sigma_{\bar{P}_1}^2 + \sigma_{\bar{P}_2}^2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

If sample size n_1 and n_2 are large, that is, $n_1 \geq 30$ and $n_2 \geq 30$, then the sampling distribution of difference of proportions is closely approximated by a normal distribution.

Illustration 1

Ten per cent of machines produced by company A are defective and five per cent of the those produced by company B are defective. A random sample of 250 machines is taken from company A and a random sample of 300 machines from company B. What is the probability that the difference in sample proportion is less than or equal to 0.02?

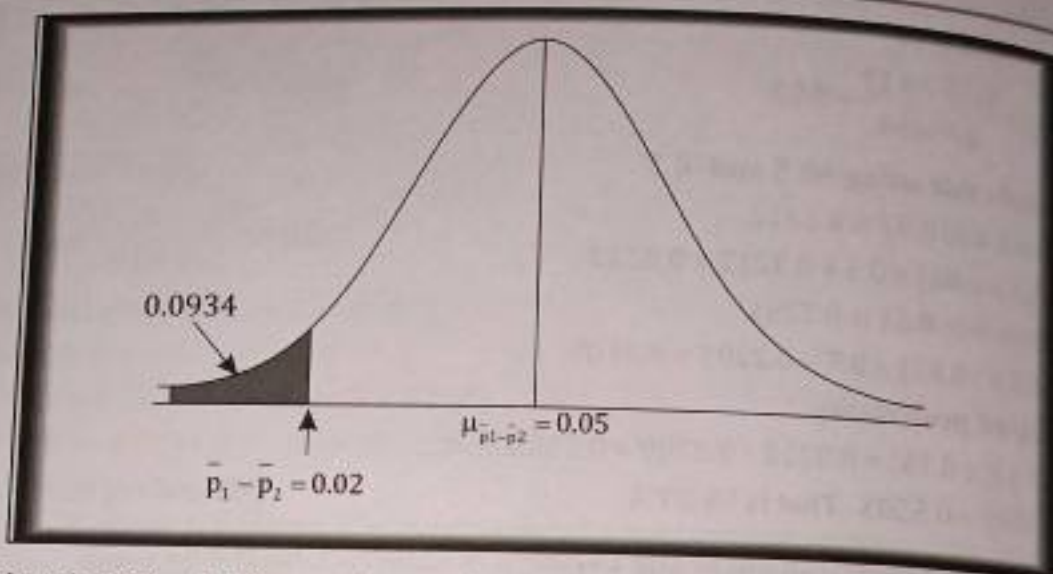
Solution

We are given the following information

$$\mu_{\bar{P}_1 - \bar{P}_2} = \mu_{\bar{P}_1} - \mu_{\bar{P}_2} = p_1 - p_2 = 0.10 - 0.05 = 0.05; n_1 = 250 \text{ and } n_2 = 300$$

Thus standard error of the difference in a sample proportion is given by

$$\sigma_{\bar{P}_1 - \bar{P}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}} = \sqrt{\frac{0.10 \times 0.90}{250} + \frac{0.05 \times 0.95}{300}} = \sqrt{\frac{0.90}{250} + \frac{0.0475}{300}} = \sqrt{0.00052} = 0.0228$$



The desired probability of difference in sample proportion is given by

$$P[(\bar{p}_1 - \bar{p}_2) \leq 0.02] = P\left[z \leq \frac{(\bar{p}_1 - \bar{p}_2) - (\mu_{\bar{p}_1 - \bar{p}_2})}{\sigma_{\bar{p}_1 - \bar{p}_2}}\right] = P\left[z \leq \frac{0.02 - 0.05}{0.0228}\right]$$

$$= P[z \leq -1.32] = 0.5000 + 0.4066 = 0.0934$$

Hence, the desired probability for the difference in sample proportion is 0.0934.

Illustration 2

Two independent populations are studied.

Population 1 has proportion $P_1 = 0.60$. Sample size $n_1 = 200$.

Population 2 has proportion $P_2 = 0.50$. Sample size $n_2 = 150$.

Find $P(0.05 \leq \bar{P}_1 - \bar{P}_2 \leq 0.15)$

Solution

Step 1 : Write the formula

$$z = \frac{(\bar{P}_1 - \bar{P}_2) - (\mu_{\bar{P}_1 - \bar{P}_2})}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}}$$

Mean of the difference

$$\mu_{\bar{P}_1 - \bar{P}_2} = P_1 - P_2$$

Standard error of the difference

$$\sigma_{\bar{P}_1 - \bar{P}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

Step 2: Substitute values

$$\mu_{\bar{p}_1 - \bar{p}_2} = 0.60 - 0.50 = 0.10$$

$$\sigma_{\bar{p}_1 - \bar{p}_2} = \sqrt{\frac{0.60 \cdot 0.40}{200} + \frac{0.50 \cdot 0.50}{150}} = \sqrt{\frac{0.24}{200} + \frac{0.25}{150}} = \sqrt{0.0012 + 0.0016667} = \sqrt{0.0028667} = 0.0535$$

Step 3: Convert bounds to z

Lower bound $a = 0.05$

$$z_1 = \frac{0.05 - 0.10}{0.0535} = -0.93$$

Upper bound $b = 0.15$

$$z_2 = \frac{0.15 - 0.10}{0.0535} = 0.93$$

Step 4: Normal table with +0.5 and -0.5

Area from 0 to 0.93 is 0.3238.

Hence

$$P(Z \leq 0.93) = 0.5 + 0.3238 = 0.8238$$

$$P(Z \leq -0.93) = 0.5 - 0.3238 = 0.1762$$

Step 5: Required probability

$$P(0.05 \leq \bar{p}_1 - \bar{p}_2 \leq 0.15) = 0.8238 - 0.1762 = 0.6476$$

Probability ≈ 0.6476 . That is 64.76%

Illustration 3

Two independent customer surveys are run.

Population 1 has true proportion $p_1 = 0.55$. Sample size $n_1 = 240$.

Population 2 has true proportion $p_2 = 0.48$. Sample size $n_2 = 200$.

Find $P(\bar{p}_1 - \bar{p}_2 > 0.12)$

Solution

Step 1: Formula

$$z = \frac{(\bar{p}_1 - \bar{p}_2) - (p_1 - p_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}}$$

Mean of the difference

$$\mu_{\bar{p}_1 - \bar{p}_2} = p_1 - p_2$$

Standard error of the difference

$$\sigma_{\bar{p}_1 - \bar{p}_2} = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

Step 2: Substitute values

$$\mu_{\bar{P}_1 - \bar{P}_2} = 0.55 - 0.48 = 0.07$$

$$\sigma_{\bar{P}_1 - \bar{P}_2} = \sqrt{\frac{0.55 \cdot 0.45}{240} + \frac{0.48 \cdot 0.52}{200}} = \sqrt{\frac{0.2475}{240} + \frac{0.2496}{200}}$$

$$= \sqrt{0.00103125 + 0.001248} = \sqrt{0.00227925} = 0.04774$$

Step 3: Convert the bound to z

Target difference equals to 0.12

Center at 0.07

$$z = \frac{0.12 - 0.07}{0.04774} = \frac{0.05}{0.04774} \approx 1.05$$

Step 4: Use Normal table with +0.5

Tables often give the area from 0 to z

For $z = 1.05$, the area from 0 to z is 0.3531.

$$P(Z \leq 1.05) = 0.5 + 0.3531 = 0.8531$$

Right tail

$$P(Z > 1.05) = 1 - 0.8531 = 0.1469$$

Step 5: Conclusion

$$P(\bar{P}_1 - \bar{P}_2 > 0.12) \approx 0.1469$$

So the probability is about 14.69%

9.8 Standard Error

The standard deviation of the sampling distribution of a statistic is called the **standard error** of the statistic. The standard deviation of the distribution of the sample means is called the **standard error of the mean**. Likewise, the standard deviation of the distribution of the sample proportions is called the **standard error of the proportion**.

The standard error is popularly known as **sampling error**. Sampling error throws light on the precision and accuracy of the estimate. The standard error is inversely proportional to the sample size i.e. the larger the sample size the smaller the standard error.

Standard error measures the dispersion of all possible values of a statistic from repeated samples of fixed size drawn from a given population. It is used to set confidence limits for population parameters in tests of significance. The standard error of $\bar{X}$ is used to find confidence limits for the population mean μ . The standard error of $\bar{P}$ is used to find confidence limits for the population proportion P .

Statistic	Standard error	Remarks
Sample Mean	$\frac{\sigma}{\sqrt{n}}$	Population size is infinite or sample with replacement
	$\frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$	Population size N finite or sample without replacement

Sample Proportion	$\sqrt{\frac{pq}{n}}$	Population size is infinite or sample with replacement.
	$\sqrt{\frac{pq}{n}} \sqrt{\frac{N-n}{N-1}}$	Population size N finite or sample without replacement.

Illustration 1

A process has population standard deviation $\sigma_x = 10$ units. A simple random sample of size $n = 25$ is taken. Assume the population is effectively infinite or sampling is with replacement. Find the standard error mean.

Solution

Given values : $\sigma_x = 10$ $n = 25$

$$\text{Formula : } \sigma_{\bar{x}} = \frac{\sigma_x}{\sqrt{n}}$$

Substitution :

$$\sigma_{\bar{x}} = \frac{10}{\sqrt{25}} = \frac{10}{5} = 2$$

$\sigma_{\bar{x}} = 2$ units.

Illustration 2

A machine output has population standard deviation $\sigma_x = 24$ minutes. A sample of size $n = 64$ is drawn. Assume the population is infinite or sampling is with replacement. Find the standard error of the mean.

Solution

Given values : $\sigma_x = 24$ and $n = 64$

$$\text{Formula } \sigma_{\bar{x}} = \frac{\sigma_x}{\sqrt{n}}$$

Substitution :

$$\sigma_{\bar{x}} = \frac{24}{\sqrt{64}} = \frac{24}{8} = 3$$

$\sigma_{\bar{x}} = 3$ minutes.

Illustration 3

A warehouse contains a finite lot of size $N = 400$. The population standard deviation is $\sigma_x = 15$ kilograms. A simple random sample of size $n = 40$ is taken without replacement. Find the standard error of the mean using the finite population correction.

Solution

Given values : $N = 400$ $\sigma_x = 15$ $n = 40$

$$\text{Formula } \sigma_{\bar{x}} = \frac{\sigma_x}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

Substitution

$$\text{Base term } \frac{15}{\sqrt{40}} = 2.372$$

$$\text{Finite population correction } \sqrt{\frac{400-40}{400-1}} = \sqrt{\frac{360}{399}} \approx 0.949$$

$$\begin{aligned} \text{Standard error } \sigma_{\bar{x}} &= 2.372 \times 0.949 \approx 2.25 \\ \sigma_{\bar{x}} &\approx 2.25 \text{ kilograms.} \end{aligned}$$

Illustration 4

A quality lot has finite size $N = 120$. The population standard deviation is $\sigma_x = 6\text{cm}$. A simple random sample of size $n = 30$ is taken without replacement. Find the standard error of the mean using the finite population correction.

Solution

Given values : $N = 120$ $\sigma_x = 6$ $n = 30$

$$\text{Formula } \sigma_{\bar{x}} = \frac{\sigma_x}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}$$

Substitution

$$\text{Base term } \frac{6}{\sqrt{30}} = 1.095$$

$$\text{Finite population correction } \sqrt{\frac{120-30}{120-1}} = \sqrt{\frac{90}{119}} \approx 0.870$$

$$\begin{aligned} \text{Standard error } \sigma_{\bar{x}} &= 1.095 \times 0.870 \approx 0.952 \\ \sigma_{\bar{x}} &\approx 0.952 \text{ c.m.} \end{aligned}$$

Illustration 5

A survey is drawn from an effectively infinite population. The true proportion is $p = 0.40$. The sample size is $n = 200$. Find the standard error of the sample proportion.

Solution

Given values : $p = 0.40$ $q = 0.60$ $n = 200$

Formula

$$\sigma_{\bar{p}} = \sqrt{\frac{pq}{n}}$$

$$\sigma_p = \sqrt{\frac{0.40 \times 0.60}{200}} = \sqrt{\frac{0.24}{200}} = \sqrt{0.0012} = 0.0346$$

$$\sigma_p \approx 0.0346$$

Illustration 6

A poll is taken from an infinite population. The true proportion is $p = 0.12$. The sample size is $n = 500$. Find the standard error of the sample proportion.

Solution

Given values: $p = 0.12$ $q = 0.88$ $n = 500$

Formula

$$\sigma_p = \sqrt{\frac{pq}{n}}$$

$$\sigma_p = \sqrt{\frac{0.12 \times 0.88}{500}} = \sqrt{\frac{0.1056}{500}} = \sqrt{0.0002112} = 0.0145$$

$$\sigma_p \approx 0.0145$$

Illustration 7

A quality check is done on a finite lot of size $N = 1000$. The population is $p = 0.30$. A simple random sample of size $n = 120$ is drawn without replacement. Find the standard error using the finite population correction.

Solution

Given values: $N = 1000$ $p = 0.30$ $q = 0.70$ $n = 120$.

Formula

$$\sigma_p = \sqrt{\frac{pq}{n}} \sqrt{\frac{N-n}{N-1}}$$

Substitution:

Base term

$$\sqrt{\frac{0.30 \times 0.70}{120}} = \sqrt{\frac{0.21}{120}} = \sqrt{0.00175} = 0.0418$$

Finite population correction

$$\sqrt{\frac{1000-120}{1000-1}} = \sqrt{\frac{880}{999}} = 0.9397$$

Standard error

$$\sigma_p = 0.0418 \times 0.9397 \approx 0.0393$$

$$\sigma_p \approx 0.0393$$

Illustration 8

A study inspects a finite warehouse stock of size $N = 250$. The population proportion is $p = 0.15$. A sample of size $n = 80$ is taken without replacement. Find the standard error using the finite population correction

Solution

Given values : $N = 250$ $p = 0.15$ $q = 0.85$ $n = 80$.

Formula:

$$\sigma_{\bar{p}} = \sqrt{\frac{pq}{n}} \sqrt{\frac{N-n}{N-1}}$$

Substitution:

Base term

$$\sqrt{\frac{0.15 \times 0.85}{80}} = \sqrt{\frac{0.1275}{80}} = \sqrt{0.00159375} = 0.0399$$

Finite population correction

$$\sqrt{\frac{250-80}{250-1}} = \sqrt{\frac{170}{249}} = 0.8264$$

Standard error $\sigma_{\bar{p}} = 0.0399 \times 0.8264 \approx 0.0330$

$$\sigma_{\bar{p}} = 0.0330$$

Illustration 9

A population consists of the five numbers 2, 3, 6, 8, 11. Consider all possible samples of size 2 which can be drawn with replacement from this population. Find (i) mean of the population (ii) the standard deviation of the population (iii) the mean of the sample distribution of means and (iv) the standard error of means.

Solution

(i) Mean of the population

$$\mu_x = \frac{2+3+6+8+11}{5} = \frac{30}{5} = 6$$

Population mean equals 6.

(ii) Standard deviation of the population

Use population variance. Divide by N

$$\sigma_x^2 = \frac{1}{5} [(2-6)^2 + (3-6)^2 + (6-6)^2 + (8-6)^2 + (11-6)^2]$$

Squared deviations 16 and 9 and 0 and 4 and 25

Sum equals 54.

$$\sigma_x^2 = \frac{54}{5} = 10.8 \Rightarrow \sigma_x = \sqrt{10.8} = 3.286$$

Population standard deviation equals about 3.286.

(iii) **Mean of the sampling distribution of sample means**

Sampling is with replacement.

$$E[\bar{X}] = \mu_{\bar{X}} = \mu_x = 6$$

Mean of the sampling distribution equals 6.

(iv) **Standard error of the sample means**

Sampling is with replacement.

$$\text{Use } \sigma_{\bar{x}} = \frac{\sigma_x}{\sqrt{n}}$$

Apply the rounded value as requested

$$\sigma_{\bar{x}} = \frac{3.286}{\sqrt{2}} = \frac{3.286}{1.4142} = 2.324$$

Standard error of the means equals about 2.324.

Illustration 10

Assume that the monthly savings of 1000 employees working in a factory are normally distributed with mean Rs. 2000 and standard deviation Rs. 50. If 25 samples consisting of 4 employees each are obtained, what would be the mean and standard deviation of the resulting sampling distribution of means if sampling were done (i) with replacement, (ii) without replacement.

Solution

Given: $N = 1000$, $\mu = 2000$, $\sigma = 50$, $n = 4$

i. $\mu_{\bar{x}} = \mu = 2000$

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{50}{\sqrt{4}} = 25$$

ii. Sampling without replacement

The mean of the sampling distribution of the means is

$$\mu_{\bar{x}} = \mu = 2000$$

The standard deviation of the sampling distribution of means is

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} = \frac{50}{\sqrt{4}} \sqrt{\frac{1000-4}{1000-1}}$$

$$= (25) \sqrt{\frac{996}{999}} = 25(\sqrt{996}) = (25)(0.9984) = 24.96$$

9.9 Review Questions**Short Answer Type Questions**

1. Write the meaning of sampling in statistics.
2. Write the meaning of population in statistics.
3. Write the meaning of the census method.
4. State any two merits of the census method.
5. State any two limitations of the census method.
6. Write the meaning of a finite population with two examples.
7. Write the meaning of an infinite population with two examples.
8. Define parameter and give two examples.
9. Define statistic and give two examples.
10. Write the meaning of simple random sampling and give one example.
11. Write the meaning of systematic sampling.
12. Write the meaning of stratified sampling.
13. Write the meaning of cluster sampling.
14. Write the meaning of convenience sampling.
15. Write the meaning of judgement sampling.
16. Write the meaning of quota sampling.
17. Write the meaning of snowball sampling.
18. State the principle of statistical regularity in simple words.
19. State the principle of inertia of large numbers.
20. State the principle of validity.
21. State the principle of optimisation.
22. Write the meaning of sampling with replacement. Give an example.
23. Write the meaning of sampling without replacement. Give an example.
24. Define sampling distribution.
25. State the central limit theorem.
26. What do you mean by Sampling Distribution of Sample Mean? Give an example.
27. What do you mean by Sampling Distribution of Sample Proportion? Give an example.
28. Write how sample size n affects the standard error.
29. State how sample size affects the spread of the sampling distribution.
30. State the property that relates the average of all sample means to the population mean.
31. Write the meaning of standard error of the mean.
32. State what happens to the standard error when sample size increases.

33. Write the meaning of standard error.
34. Write the meaning of a sample mean and sample proportion.
35. State when a normal approximation is reasonable for proportions.
36. Write any two differences between census and sampling.
37. State one situation where census is preferred and state one situation where sampling is preferred.
38. A district has 85,000 households. A survey selects $n=900$ households to study LPG usage. Identify the population, the sample, and the sampling unit.
39. A company produced 18,000 fans in a month. A quality team inspects $n=240$ fans. State whether the population is finite or infinite and justify.
40. Population $\{5, 7, 9, 11\}$. List all unordered samples of size $n=2$ without replacement.
41. The population standard deviation is $\sigma=30$ g. Find $\sigma_{\bar{x}}$ for $n=25$ and $n=100$. State which is more precise.
42. The population standard deviation is $\sigma=18$. Find $\sigma_{\bar{x}}$ for $n=9$ and $n=81$.

Long Answer Type Questions

1. Explain the concept of population and sample with examples.
2. Explain the concept of parameter and statistic with examples.
3. Describe how a sample statistic helps to learn about a population parameter.
4. Differentiate between Parameter and Statistic.
5. Describe the census method and the sampling method. Write detailed merits and limitations of each.
6. Explain probability sampling methods with examples.
7. Explain non-probability sampling methods with examples.
8. Differentiate between Census and Sampling.
9. What is Sampling? Explain the need for sampling.
10. Write the advantages and disadvantages of sampling.
11. Explain Sampling with and without Replacement with examples.
12. State and explain the principles of sampling.
13. Explain the meaning of sampling distribution in simple words. Write the objectives of Sampling Distribution.
14. What is Normal vs Non-Normal Population Distributions? Explain with an example.
15. Explain the Properties of Sampling Distribution.
16. State the central limit theorem. Explain what happens to the shape of the sampling distribution as the sample size grows. Explain with an example.
17. Write the properties of the sampling distribution of the sample mean.
18. The ages (in years) of three employees in a small shop are : $\{20, 30, 40\}$. Take all possible samples of size 2 with replacement. Construct the sampling distribution of the sample mean and verify that the mean of the sampling distribution equals the population mean.

19. Population values : {20, 30, 40}. Sampling is with replacement and Sample size $n = 3$.
 1. Construct the sampling distribution of the sample mean.
 2. Find the $E[\bar{X}]$ and verify it equals the population mean.
20. Population values are {4, 6, 8, 10}. Sample size is $n=2$ and Sampling is without replacement.
 1. Construct the sampling distribution table of the sample mean.
 2. Find the mean of this distribution. Verify that it equals the population mean.
21. Population values are {4, 6, 8, 10, 12}. Sample size is $n=3$. Sampling is without replacement.
 1. Construct the sampling distribution table of the sample mean.
 2. Find the mean of this distribution. Verify that it equals the population mean.
22. In a certain locality, the average rent paid by all tenants amounts to ₹2000 pm with a standard deviation of ₹450. However, the population distribution of rents pertaining to all tenants in that city is positively skewed. Find out the mean and standard deviation of $\bar{X}$ when the sample size is (i) 30 and (ii) 100. Also describe the shape of its sampling distribution in both cases.
23. In a large city, individual monthly electricity bills are right-skewed (many small bills, a few very large ones). The population mean bill is Rs.3,400 with population standard deviation Rs. 1,000. If we draw random samples of sizes $n = 25$ and $n = 100$, find the mean and standard deviation of the sample mean $\bar{X}$ for each case. Then state what this implies about the sampling distribution as n grows.
24. Delivery times for online orders in a city are right-skewed (most arrive quickly, a few take much longer). The population mean delivery time is $\mu = 3.0$ days and the population standard deviation is $\sigma = 1.5$ days. If we draw random samples of sizes $n = 16$ and $n = 64$, find the mean and standard deviation of the sample mean $\bar{X}$ for each case. Then state what this implies as n grows.
25. A factory packs 1 kg rice packets. The weight of each packet follows a Normal distribution with:
 - Mean $\mu = 500$ grams
 - Standard deviation $\sigma = 10$ grams
 A sample of size $n = 36$ is taken.
 1. What is the mean and standard deviation of the sampling distribution of sample mean?
 2. Find the probability that sample mean is between 495 and 505 grams.
 3. Find the probability that sample mean is greater than 505 grams.
26. In a normally distributed population the average income per household is Rs 10000 per month. The population standard deviation is Rs 800. A random sample of size $n=100$ households is taken. Find the probability that the sample mean income lies between Rs 9800 and Rs 10100.
27. A machine fills packets. Packet weights are normally distributed. Mean $\mu = 50$ g. Known standard deviation $\sigma = 12$ g and Sample size $n = 25$. Find $P(\bar{X} < 47)$.

28. Packet weights follow a Normal distribution.

Mean $\mu = 205$ g.

Standard deviation $\sigma = 25$ g.

Sample size $n = 16$.

Find $P(\bar{X} > 210)$.

29. Packet weights follow a Normal distribution.

Mean $\mu = 75$ g.

Standard deviation $\sigma = 15$ g.

Sample size $n = 36$.

Find $P(\bar{X} < 72)$.

30. A corporation tracks staff turnover across 40 branches of equal size. The population standard deviation of annual turnover is 80 employees. A sample of 4 branches is selected without replacement. Find the standard error of the mean.

31. The mean length of life of a certain cutting tool is 42 hours with a standard deviation of 2.5 hours. What is the probability that a simple random sample of size 50 drawn from this population will have a mean between 41 hours and 43 hours?

32. Two smartphone brands are tested for battery life.

Population 1. Mean $\mu_1 = 310$ hours. Standard deviation $\sigma_1 = 40$ hours.

Population 2. Mean $\mu_2 = 300$ hours. Standard deviation $\sigma_2 = 50$ hours.

Independent random samples are taken. Sample sizes are $n_1 = 64$ and $n_2 = 49$.

Find $P((\bar{X}_1 - \bar{X}_2) > 15)$.

33. Two independent populations are considered.

Population 1. $\mu_1 = 82$. $\sigma_1 = 12$. Sample size $n_1 = 36$.

Population 2. $\mu_2 = 78$. $\sigma_2 = 15$. Sample size $n_2 = 25$.

Find $P(2 \leq \bar{X}_1 - \bar{X}_2 \leq 8)$.

34. A manufacturer of watches has determined from experience that 3 per cent of the watches he produces are defective. If a random sample of 400 watches is examined, what is the probability that the proportion defective is between 0.02 and 0.035?

35. A Factory produces items with defect rate $p = 0.20$. A random sample of $n = 250$ items is inspected. Find $P(0.18 \leq \bar{P} \leq 0.25)$.

36. A Factory has a small finite lot of $N = 200$ items. The population defect proportion is $p = 0.16$. A simple random sample of $n = 50$ items is taken without replacement.

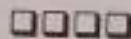
Find $P(0.12 \leq \bar{P} \leq 0.20)$.

37. A Ware house has a finite stock of $N = 400$ items. The true defect proportion is $p = 0.12$.

A simple random sample of $n = 80$ items is drawn without replacement.

Find $P(0.10 \leq \bar{P} \leq 0.14)$.

38. Ten per cent of machines produced by company A are defective and five per cent of the those produced by company B are defective. A random sample of 250 machines is taken from company A and a random sample of 300 machines from company B. What is the probability that the difference in sample proportion is less than or equal to 0.02?
39. Two independent populations are studied.
 Population 1 has proportion $P_1 = 0.60$. Sample size $n_1 = 200$.
 Population 2 has proportion $P_2 = 0.50$. Sample size $n_2 = 150$.
 Find $P(0.05 \leq \bar{P}_1 - \bar{P}_2 \leq 0.15)$
40. Two independent customer surveys are run.
 Population 1 has true proportion $p_1 = 0.55$. Sample size $n_1 = 240$.
 Population 2 has true proportion $p_2 = 0.48$. Sample size $n_2 = 200$.
 Find $P(\bar{P}_1 - \bar{P}_2 > 0.12)$
41. A process has population standard deviation $\sigma_x = 20$ units. A simple random sample of size $n = 50$ is taken. Assume the population is effectively infinite or sampling is with replacement. Find the standard error mean.
42. A machine output has population standard deviation $\sigma_x = 32$ minutes. A sample of size $n = 64$ is drawn. Assume the population is infinite or sampling is with replacement. Find the standard error of the mean.
43. A warehouse contains a finite lot of size $N = 800$. The population standard deviation is $\sigma_x = 30$ kilograms. A simple random sample of size $n = 80$ is taken without replacement. Find the standard error of the mean using the finite population correction.
44. A quality lot has finite size $N = 240$. The population standard deviation is $\sigma_x = 16$ cm. A simple random sample of size $n = 50$ is taken without replacement. Find the standard error of the mean using the finite population correction.
45. A survey is drawn from an effectively infinite population. The true proportion is $p = 0.40$. The sample size is $n = 200$. Find the standard error of the sample proportion.
46. A poll is taken from an infinite population. The true proportion is $p = 0.12$. The sample size is $n = 500$. Find the standard error of the sample proportion.
47. A quality check is done on a finite lot of size $N = 1000$. The population is $p = 0.30$. A simple random sample of size $n = 120$ is drawn without replacement. Find the standard error using the finite population correction.



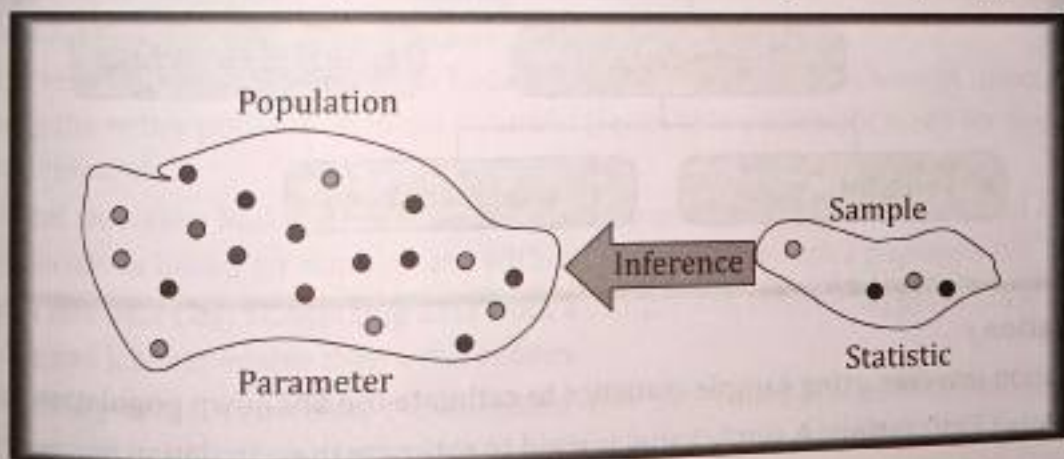
STATISTICAL INFERENCE

10.1 Statistical Inference

Statistics is broadly divided into two branches: **Descriptive Statistics** and **Inferential Statistics**.

1. **Descriptive Statistics** deals with the presentation and summarisation of data using tables, charts, and numerical measures such as mean and standard deviation.
2. **Inferential Statistics**, on the other hand, goes a step ahead. It helps to draw conclusions or generalisations about a larger group (**called population**) based on the observations from a smaller group (**called sample**).

Let us understand the concept of Inferential Statistics through the diagram given:



- **Population:** This is the larger group that contains all the individuals or items under study. For example, all students in a college, or all voters in a city. Inside the population, there are parameters. A parameter is a numerical value that describes a certain characteristic of the population such as population mean, population variance, or population proportion.
- **Sample:** This is a smaller group selected from the population. It is easier and more practical to study a sample than the whole population. Inside the sample, there are statistics. A statistic is a numerical value that describes a characteristic of the sample, such as sample mean, sample variance, or sample proportion.
- **Inference :** Between the sample and the population, the diagram shows inference. This arrow indicates the process of Statistical Inference. It is the method of using data from the sample to estimate or test characteristics of the population.



Meaning : Statistical Inference

Statistical Inference is the process of using sample data to make judgments, predictions, or decisions about the characteristics of a population. It allows the understanding of population parameters when it is not possible or practical to study the entire population.

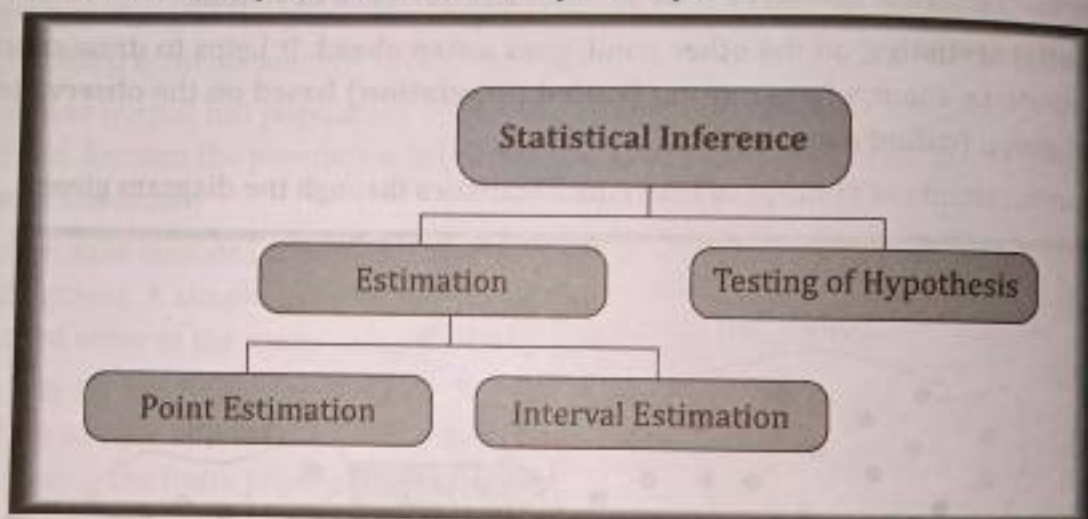
This process helps to answer important questions such as:

- What is the average income of people in a country?
- Will a new medicine be effective for the entire population?
- What is the voter preference in an upcoming election?

But instead of asking everyone, a small sample is studied. Based on that, conclusions are drawn.

Main Aspects of Statistical Inference

Statistical inference is mainly divided into two important parts:



1. Estimation :

Estimation involves using sample statistics to estimate the unknown population parameters.

- **Point Estimation:** A single value is used to estimate the population parameter.

Example: Using sample mean ($\bar{X}$) to estimate population mean (μ).

- **Interval Estimation:** A range of values (confidence interval) is given, within which the parameter is expected to lie.

Example: Estimating that population mean (μ) lies between 50 and 58 with 95% confidence.

2. Testing of Hypothesis :

- Hypothesis Testing is a procedure used to test whether a statement about a population parameter is true or not based on sample data.
- **Example:** A company claims that the average life of its battery is 400 hours. A sample of batteries is tested. If the average comes out significantly different, the claim may be rejected.

10.2 Estimation

Estimation is a statistical process that uses sample statistics to make inferences about population parameters. The purpose is to provide an educated guess of values such as the population mean, proportion, or variance. It focuses on finding a likely range within which the true parameter lies, along with a measure of precision and confidence.

This process plays a vital role in decision making. It allows researchers and analysts to draw valid conclusions about a population without examining every individual. Instead, smaller groups or samples are studied, and their statistics such as the sample mean, sample proportion, or sample standard deviation are used as the basis for inference.

In practice, estimation is often supported by statistical techniques like confidence intervals and hypothesis testing. Confidence intervals provide a range within which the parameter is expected to fall with a certain degree of confidence, while hypothesis testing helps in verifying assumptions or claims about the population.



Definition : Estimation

Estimation in statistics refers to the process of using sample data to infer the value of a population parameter. Its main goal is to identify a likely range for the parameter and provide a measure of precision along with a level of confidence in the estimate.

10.2.1 Importance of Estimation

Estimation is very important in statistics because it helps to make conclusions about a population when studying the entire population is not possible. It provides a scientific basis for decision making, planning, and research.

- 1. Practical Decision Making :** Estimation allows businesses, governments, and researchers to make decisions based on sample data without studying the entire population.
- 2. Saves Time and Cost :** Collecting data from a sample and estimating population parameters is quicker and less expensive than a full census.
- 3. Forecasting and Prediction :** Businesses use estimation to predict sales and demand. Governments use it to forecast population growth, inflation, and employment.
- 4. Policy Formulation :** Estimation helps policymakers in designing economic, social, and health-related policies based on estimated figures like GDP or literacy rate.
- 5. Reliability and Confidence :** Estimation gives not just a single value but also a range with confidence levels, which shows the reliability of the results.
- 6. Foundation for Hypothesis Testing :** Hypothesis testing begins with estimation, which provides the basic values needed to test assumptions about a population.

10.2.2 Properties of a Good Estimator

A sample statistic which is used to estimate a population parameter is known as **estimator**. A good estimator is one which is as close to the true value of population parameter as possible. A good estimator possesses the following properties :

i. Unbiasedness : An estimator is unbiased when its expected value equals the true parameter.
Example : A college samples marks of $n=50$ students from a large class. The average of the 50 marks estimates the true class mean. On repeated sampling the average of all such sample means equals the true mean. Hence $\bar{X}$ is unbiased.

ii. Consistency : An estimator is said to be consistent if the estimate tends to approach the parameter as the sample size increases.

Example : Estimate the average monthly mobile bill in a city. Take a sample of 100 customers and then 10,000 customers. The 10,000 customer average will be closer to the true city average.

iii. Efficiency : If we have two unbiased estimators for the same population parameter, the first estimator is said to be more efficient than the second estimator if the standard error of the first estimator is smaller than that of the second estimator for the same sample size.

Example : For estimating the average monthly electricity bill of households in Bengaluru, both the sample mean and the sample median can be used as estimators of the true population mean. However, the sample mean has a smaller variance compared to the sample median when the data is approximately normal. This means the sample mean produces values closer to the actual population mean more frequently than the median. Hence, the sample mean is considered a more efficient estimator than the sample median in this case.

iv. Sufficiency : If an estimator possesses all information regarding the parameter, then the estimator is said to be a sufficient estimator.

Example : In a school survey each student marks 1 for having a bus pass and 0 for not having it. The aim is to estimate the true proportion p of students with a bus pass. The total number of 1s in the sample denoted by T is sufficient for p . Once T is known the order of the 0s and 1s does not matter and no extra information is gained. Hence T is a sufficient statistic for the population proportion.

10.2.3 Types of Estimation

There are two main types of estimation:

1. Point Estimation

Point estimation is a statistical method that gives a single number as the best guess of a population parameter. Since it is not practical to study an entire population, a smaller sample is taken. A statistic such as the sample mean, proportion, or variance is then calculated from the sample. This single value is called the point estimate of the true population parameter.

The main aim of point estimation is to provide a value that is as close as possible to the true parameter. However, because samples vary, the point estimate may differ from the actual population value. This difference is called sampling error. To understand the reliability of a point estimate, statisticians use measures such as the standard error, which tells how much the estimate would vary if different samples were taken.

A good point estimator should have certain desirable properties. It should be unbiased, meaning that on average it equals the true parameter. It should be consistent, so that as the sample size grows the estimate gets closer to the true value. It should also be efficient, which means it has the smallest possible standard error among competing estimators.

Example 1 Estimating a Population Mean

Suppose we want to find the average height of all students in a school. Instead of measuring every student, a random sample of 50 students is taken. Their average height is found to be 170 cm. This value, 170 cm, is used as the point estimate of the true population mean height. Although it may not exactly match the actual average of the whole school, it serves as a close estimate.

Example 2 Estimating a Population Proportion

Suppose in an election survey we want to estimate the proportion of voters supporting a candidate. A random sample of 500 voters is taken, and 300 of them say they support the candidate. The sample proportion is $(300/500 = 0.60)$ or 60%. This single value is the point estimate of the true proportion of support in the entire voter population.

2. Interval Estimation

Interval estimation is a statistical method used to estimate a population parameter by providing a range of values within which the true parameter is likely to fall, along with a measure of confidence in the estimate. This range of values is known as a confidence interval.

When conducting interval estimation, a point estimate (such as the sample mean or proportion) is calculated from the sample data, and then a confidence interval is constructed around this point estimate. The confidence interval provides a lower and upper bound within which the true population parameter is expected to lie with a certain level of confidence, typically expressed as a percentage (example, 95% confidence interval).

For example, if we want to estimate the average response time of a customer service team, we might calculate a 95% confidence interval for the population mean response time. This interval might be, for instance, 3 to 5 minutes. This means that we are 95% confident that the true population mean response time falls within this range.

The width of the confidence interval is influenced by factors such as the sample size and the variability of the data. A larger sample size generally leads to a narrower confidence interval, providing a more precise estimate of the population parameter.

Interval estimation is valuable because it not only provides an estimate of the population parameter but also quantifies the uncertainty associated with the estimate. It acknowledges the variability inherent in sampling and provides a more comprehensive understanding of the likely range of values for the population parameter, allowing decision-makers to make informed judgments based on the available sample data.

A confidence level shows how often the method will capture the true value in repeated samples. A 95 percent confidence level means that if the same study is repeated many times then about 95 out of 100 such intervals would contain the true parameter. It does not claim a 95 percent chance for one specific interval. Higher confidence needs a wider interval. Lower confidence gives a narrower interval. The width also depends on sample size. Larger samples give tighter intervals under the same level. The statement is valid only under the study's assumptions and sampling plan.

Example 1 Estimating a Population Mean

Suppose we want to know the average monthly pocket money of college students. A random sample of 40 students is taken, and their average comes out to ₹2,000. Instead of reporting just this single number, we can give a range, say from ₹1,900 to ₹2,100, within which the true average pocket money of all students is likely to lie with 95% confidence.

Example 2 Estimating a Population Proportion

Suppose a company wants to estimate how many customers prefer online shopping. A survey of 600 people is conducted, and 65% of them say they prefer online shopping. To show the reliability of this estimate, the company reports that the true proportion of customers who prefer online shopping is likely to fall between 62% and 68% with 95% confidence.

10.3 Hypothesis Testing

In research and statistics, it is often not enough to only describe data. Many times, decisions must be made or claims must be tested. For example, a company may claim that its new battery lasts longer than the old one, or a school may want to check if the average marks of students have improved after introducing a new teaching method. In such situations, hypothesis testing is used.

Hypothesis testing provides a systematic way of checking whether the information from a sample supports or contradicts a belief or assumption about a population. It works on the idea that we begin with an initial assumption, called the null hypothesis, and then check if there is enough evidence in the sample data to reject it in favor of an alternative explanation, called the alternative hypothesis.

Scenarios where hypothesis testing is applied:

- A hospital tests if a new drug is more effective than the existing medicine.
- An education board tests if online classes have changed the average exam scores of students.
- An election survey checks if support for a candidate is greater than 50%.
- A factory tests if the defect rate in its production is within an acceptable limit.

**Definitions : Hypothesis Testing**

1. Hypothesis testing is a statistical method to check a claim about a population using sample data. It starts with a default claim called the null hypothesis. It also sets an alternative claim called the alternative hypothesis. The sample evidence is compared with the null hypothesis to see if the difference is small or large. If the evidence is strong then the null hypothesis is rejected. If the evidence is weak then the null hypothesis is not rejected. This gives a clear decision based on data and a chosen level of significance.
2. Hypothesis testing is a statistical method used to test a claim or assertion about a population parameter based on sample data.
3. Hypothesis testing involves using sample data to make inferences about the population parameter, allowing us to assess whether the presumption about the population is supported by the sample evidence.

Example 1 Testing a Mean

Start with the claim that the average daily income in Mangaluru is ₹50. This is the null hypothesis. Take a random sample of residents and find the sample mean. If the sample mean is very close to ₹50 then the claim is accepted. If the sample mean is far from ₹50 then the claim is rejected. For instance a sample mean of ₹50 leads to claim accepted. A sample mean around ₹55 leads to claim rejected.

Example 2 Average Marks of Students

A college believes that the average marks of its students are 60. A random sample of students is taken and their average is compared with 60. If the difference is too large, the belief may be rejected, and it may be concluded that the true average marks are different.

Example 3 Effectiveness of a Medicine

A pharmaceutical company claims that a new medicine cures fever within 5 days on average. Doctors test this claim by giving the medicine to a group of patients and recording the recovery time. If the average recovery time is very different from 5 days, the claim may be rejected.

In simple words, hypothesis testing is a decision-making tool in statistics. It uses sample data to decide whether to accept or reject a claim about a population.

10.3.1 Importance of Hypothesis Testing

Hypothesis testing helps to make decisions from sample data. It checks if a claim about a population is reasonable. It gives a clear rule to accept or reject a claim.

1. Supports Data-driven Decisions

Hypothesis testing helps organisations and researchers decide based on facts instead of assumptions. It provides a systematic method to accept or reject a claim using sample data.

2. Controls the Risk of Wrong Decisions

By setting a level of significance (like 5%), the risk of rejecting a true claim or accepting a false claim is controlled. This makes decisions more reliable.

3. Compares Two or More Groups

It shows whether differences between groups are real or just due to chance. For example, testing if boys and girls in a school score equally in mathematics.

4. Validates Product and Process Claims

Industries use it to check if products meet standards. For example, testing if the average weight of a cement bag is truly 50 kg as claimed.

5. Guides Policy and Planning

Governments and organisations rely on hypothesis testing to check if new policies or schemes bring real changes. This avoids decisions based only on guesswork.

6. Monitors Quality Over Time

Factories and service providers test samples at regular intervals to check if quality remains within limits. It quickly detects when the process goes off-track.

7. Works with Small Samples

It is practical when studying the whole population is not possible. Even small sample data can give meaningful results through hypothesis testing.

8. Quantifies Strength of Evidence

p-values or test statistics show how strong the evidence is against a claim. This helps in making transparent and objective decisions.

9. Separates True Effect from Random Variation

It filters out random fluctuations and confirms whether an observed effect is genuine. For example, testing if a rise in crop yield is due to new fertilizer or just chance.

10. Builds Scientific Credibility

Research studies gain trust when claims are tested and supported by statistical evidence. Hypothesis testing ensures results are not random but scientifically valid.

10.3.2 Types of Hypothesis Testing

Hypothesis testing is a central tool in statistics used to test claims or assumptions about population parameters by examining sample data. Different tests are applied depending on the type of data, the number of groups, and the information available. Below are the major types explained in simple terms:

1. T-Test

A t-test is used to check whether the mean of a sample differs from a known value or whether the means of two groups are different. It is mainly applied when the sample size is small and the population standard deviation is unknown.

Example: A college wants to test if the average marks of students after a new teaching method are higher than 60. A small sample of student scores is taken, and a t-test is applied to check if the difference is significant.

2. Z-Test

A z-test is used when the sample size is large (usually more than 30) and the population standard deviation is known. It compares a sample mean with a population mean or the means of two large independent samples.

Example: A company claims that the average life of its bulbs is 1,000 hours. A large sample of bulbs is tested, and the z-test is used to decide whether the sample data supports or rejects the company's claim.

3. Analysis of Variance (ANOVA)

ANOVA is used to compare the means of three or more groups at the same time. It checks whether the differences among the groups are statistically significant or whether they occurred by chance.

Example: An agricultural researcher tests three types of fertilizers on crop yield. ANOVA is used to find out if the average yields differ significantly across the three fertilizer groups.

4. Chi-Squared Test

The chi-squared test is applied to categorical data. It checks whether there is an association between two categorical variables or whether observed frequencies differ from expected frequencies.

Example: A marketing team wants to check if customer preference (online shopping vs. offline shopping) is related to age group (young, middle-aged, senior). A chi-squared test is applied to see if there is a significant association between age and shopping preference.

10.3.3

General Procedure of Hypothesis Testing

In hypothesis testing, the goal is to decide whether to accept or reject a claim about a population parameter using sample data. The procedure is systematic and applies to all types of hypothesis tests, whether t-test, z-test, chi-square test, or ANOVA.

• Step 1. Frame the claim

Clearly state the purpose of the test. For example, a company wants to know if the average life of its product meets the advertised claim. This defines the research problem.

• Step 2. State the hypotheses

Formulate two statements.

- ▲ The null hypothesis (H_0) expresses the default claim, such as "the mean equals 1000."
- ▲ The alternative hypothesis (H_1) represents the rival claim, such as "the mean is not equal to 1000" or "the mean is greater than 1000."

• Step 3. Select the significance level

Choose a significance level (α), commonly 5% or 1%. This fixes the risk of rejecting a true claim. A smaller α gives stricter evidence requirements.

• Step 4. Decide the test and test statistic

Select an appropriate test depending on the type of data, sample size, and assumptions. Identify the test statistic (z , t , F , or χ^2) that will be calculated from the sample.

• Step 5. Collect and prepare data

Obtain a sample through random selection. Ensure observations are independent and conditions for the chosen test are satisfied.

• Step 6. Compute the test statistic

Use the sample data to calculate the value of the chosen test statistic. This value shows how far the sample result is from the claim stated in H_0 .

• Step 7. Find the p-value or critical value

Obtain the p-value corresponding to the test statistic, or compare the statistic with a critical value from statistical tables. This step measures the strength of evidence against the null hypothesis.

• Step 8. Apply the decision rule

- ▲ If the test statistic falls into the rejection region, or the p-value is less than or equal to α , then reject H_0 and accept H_1 .

▲ If the test statistic lies outside the rejection region, or the p-value is greater than α , then accept H_0 and reject H_1 .

- **Step 9. Write the conclusion in plain words**

Translate the statistical decision into simple language. For example, "There is evidence that the new medicine works better than the old one," or "There is no evidence of a difference in average exam scores."

- **Step 10. Report essentials**

Provide details such as sample size, sample mean or proportion, test name, significance level, test statistic, and p-value. Add a confidence interval for the parameter when helpful. This makes the result transparent and credible.

Example 1 T-Test for a New Teaching Method

The aim is to test whether the average mark after the new teaching method is greater than 60. The claim is "mean > 60." The decision will be based on sample data from students who studied under the new method.

- **Step 1. Frame the claim**

State the educational question clearly. The college introduced a new method of teaching. The question is whether this method raised the average mark above the known benchmark of 60. This statement sets the direction of the test.

- **Step 2. State the hypotheses**

Write two precise statements. The null hypothesis says the mean mark equals 60. The alternative hypothesis says the mean mark is greater than 60. This is a right-tailed situation because the interest is only in improvement. These statements fix what "accepted" or "rejected" means at the end.

- **Step 3. Decide the data**

Identify the target group. Select one recent batch that actually used the new method. Draw a small random sample of student marks from that batch. Use the official marks from the same examination and the same subject to keep conditions uniform. Random selection reduces bias and gives fair evidence.

- **Step 4. Check basic conditions**

Ensure that every selected student was taught with the same method and syllabus. Ensure that each mark in the sample comes from a different student so observations stay independent. Scan the data for extreme outliers. If the sample is small then assume an approximately normal pattern or confirm it with a quick plot. The population standard deviation is not known. Hence a one-sample t-test is suitable.

- **Step 5. Fix the significance level**

Choose the risk of a false alarm before seeing the result. A 5 percent level is common in education studies. This level controls the chance of accepting the improvement claim when the true mean is still 60. A 1 percent level is stricter and needs stronger evidence. Pick one level and use it consistently.

- **Step 6. Compute the test statistic**

Calculate the sample mean from the collected marks. Calculate the sample standard deviation. Combine them with the sample size to obtain the t value for a one-sample test. This single number captures how far the sample mean sits above 60 in standard-error units.

- **Step 7. Obtain the p-value or the critical value**

Use a t table to find the p-value for the right-tailed case. The p-value tells how unusual this sample would be if the true mean were exactly 60. A small p-value means strong evidence for improvement. The same decision can be made by comparing the t value with the right-tail critical value at the chosen level.

- **Step 8. Apply the decision rule**

Compare the p-value with the significance level fixed in Step 5. If p-value is less than or equal to the level then the improvement claim is accepted and the null is rejected. If p-value is greater than the level then the improvement claim is rejected and the null is accepted. The rule is mechanical and avoids subjective bias.

- **Step 9. Write the conclusion in plain words**

Translate the statistical decision to the education context. If claim accepted then state that the new method shows evidence of raising the average above 60 for this batch. If claim rejected then state that evidence is not strong for an average above 60 and the data remain consistent with 60. Keep the language simple and precise.

- **Step 10. Report essentials for transparency**

Report the sample size and the sample mean and the sample standard deviation. Mention the test name as one-sample t-test. Mention the significance level and the p-value. Add a 95 percent confidence interval for the mean to show the likely range. Record any data checks done and any exclusions made. This helps teachers and administrators read the result with full context.

Example 2 Z-Test for a Product Claim

The aim is to test whether the true average life of a light bulb equals 1000 hours. Past records give a known population standard deviation. A large sample of bulbs will be tested. The claim is "mean = 1000 hours."

- **Step 1. Frame the claim**

A manufacturer prints 1000 hours on each bulb. The question is whether the true average life matches this target. This sets the testing goal.

- **Step 2. State the hypotheses**

Write two clear statements. The null hypothesis says the mean life equals 1000 hours. The alternative hypothesis says the mean life is different from 1000 hours. This makes it a two-tailed test. Use a one-tailed form if the interest is only in higher life or only in lower life.

- **Step 3. Decide the data**

Choose a large random sample from a fresh production lot. Record the life of every sampled bulb until failure. Keep conditions uniform during testing.

- **Step 4. Check basic conditions**

Ensure random selection and independence of observations. Confirm that a reliable historical value of population standard deviation is available. Confirm that the sample size is large so the sampling distribution of the mean is approximately normal. These points make a one-sample z-test suitable.

- **Step 5. Fix the significance level**

Choose the risk level for a wrong alarm before looking at the data. A 5 percent level is common in industry. A 1 percent level is stricter and needs stronger evidence.

- **Step 6. Compute the test statistic**

Find the sample mean from the measured lifetimes. Use the known population standard deviation and the sample size to form the z value. Software or a calculator can do this. The z value shows how far the sample mean sits from 1000 in standard error units.

- **Step 7. Obtain the p-value or the critical value**

Use a z table or software to get the p-value for the chosen tail form. For a two-tailed case check both sides. The p-value shows how unusual the sample would be if the true mean were exactly 1000 hours.

- **Step 8. Apply the decision rule**

If the p-value is less than or equal to the chosen level then the claim "mean = 1000" is rejected. If the p-value is greater than the chosen level then the claim is accepted. The same decision can be made by comparing the z value with the critical z at the chosen level.

- **Step 9. Write the conclusion in plain words**

If claim rejected. Evidence shows that the average bulb life differs from 1000 hours. If claim accepted. Evidence is not strong against the 1000 hour target and the claim stands accepted for the lot tested.

- **Step 10. Report essentials for transparency**

State the sample size and the sample mean. State the known population standard deviation. Name the test as one-sample z -test. Mention the significance level and the p-value. Add a confidence interval for the mean if needed for managers and quality teams.

10.3.4 Types of Error in Hypothesis Testing

In hypothesis testing a decision is made from sample data. Samples can differ from the population. Hence mistakes can occur. These mistakes are called errors of hypothesis testing. There are two kinds. They are Type I error and Type II error.

1. Type I Error

- A Type I error happens when the null hypothesis is true and it is rejected. It is like a fire alarm that rings when there is no fire.
- The probability of a Type I error is called alpha. It is the significance level. A common choice is 5 percent. This means there is a five percent chance of rejecting a true null hypothesis.
- **Example:** A medical screening test shows disease for a healthy person. The test rejects the statement that the person is healthy. In truth the person is healthy.

2. Type II Error

- A Type II error happens when the null hypothesis is false and it is not rejected. It is like a smoke sensor that stays silent when there is a fire.
- The probability of a Type II error is called beta. The power of a test equals one minus beta. Higher power means the test can detect a real effect more often.
- **Example:** A medical screening test shows no disease for a sick person. The test accepts the statement that the person is healthy. In truth the person is sick.

A good study balances both errors by choosing a suitable significance level and by planning for enough sample size and by considering the real world cost of a false alarm versus a missed detection.

10.4 Difference between Hypothesis Testing and Estimation

Criteria	Hypothesis Testing	Estimation
Definition	Hypothesis testing is a method that tests an assumption regarding a population parameter.	Estimation is the process of inferring the value of a population parameter based on a sample.
Purpose	The purpose is to decide the value of a population parameter based on sample data.	The aim is to guess an approximate value or range of values for a population parameter.
Types	Null Hypothesis (H_0) and Alternative Hypothesis (H_1) are kinds of hypothesis testing.	Point Estimation and Interval Estimation are kinds of estimation.
Statistical Values	In hypothesis testing, we calculate a test statistic and p-value.	In estimation, we calculate an estimate (a single value or range).
Decision	A decision has to be made to either reject or fail to reject the null hypothesis.	No explicit decision is made; an estimate is provided.
Example	If a beverage company claims that its juice pack contains 300 ml of purified water, we can apply hypothesis testing to find out if this is true based on a sample.	If we want to guess the average height of all students in a class, we could use estimation based on a small sample of pupil.

10.5 Review Questions

Short Answer Type Questions

1. What is the meaning of Statistical Inference?
2. What is the meaning of Estimation in statistics?
3. Why is estimation useful when a full census is not possible?
4. What is point estimation? Give an example.
5. What is interval estimation? Give an example.
6. What is a confidence level?
7. Write any two points that show the importance of estimation.
8. What is an estimator and what makes an estimator good?
9. What is unbiasedness of an estimator?
10. What is consistency of an estimator?
11. What is efficiency of an estimator?
12. What is sufficiency of an estimator?
13. What is the meaning of Hypothesis Testing?
14. What is the null hypothesis and alternative hypothesis? Give one example.

15. Give any two real life scenarios where hypothesis testing is applied
16. What are the main types of hypothesis testing?
17. What is a t test?
18. What is a z test?
19. What is Analysis of Variance?
20. What is the chi square test?
21. What is Type I error? Give an example.
22. What is Type II error? Give an example.
23. Write any two differences between estimation and hypothesis testing.

Long Answer Type Questions

1. What is Statistical Inference? Explain the main aspects of Statistical Inference.
2. What is Estimation? Explain the types of Estimation with example.
3. Explain the types of hypothesis testing.
4. Write the general hypothesis testing procedure with an example.
5. Differentiate between estimation and hypothesis testing.
6. Write short notes on point estimation and interval estimation.
7. Explain the properties of a good estimator.
8. Describe the importance of estimation.
9. Describe the importance of hypothesis testing.
10. Explain Type I error and Type II error.

